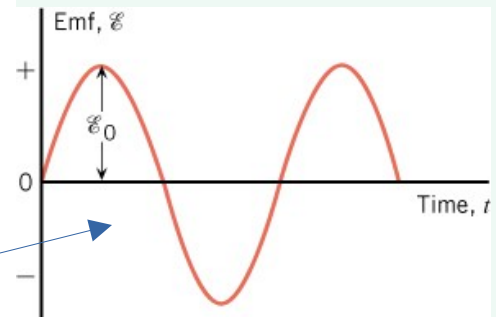
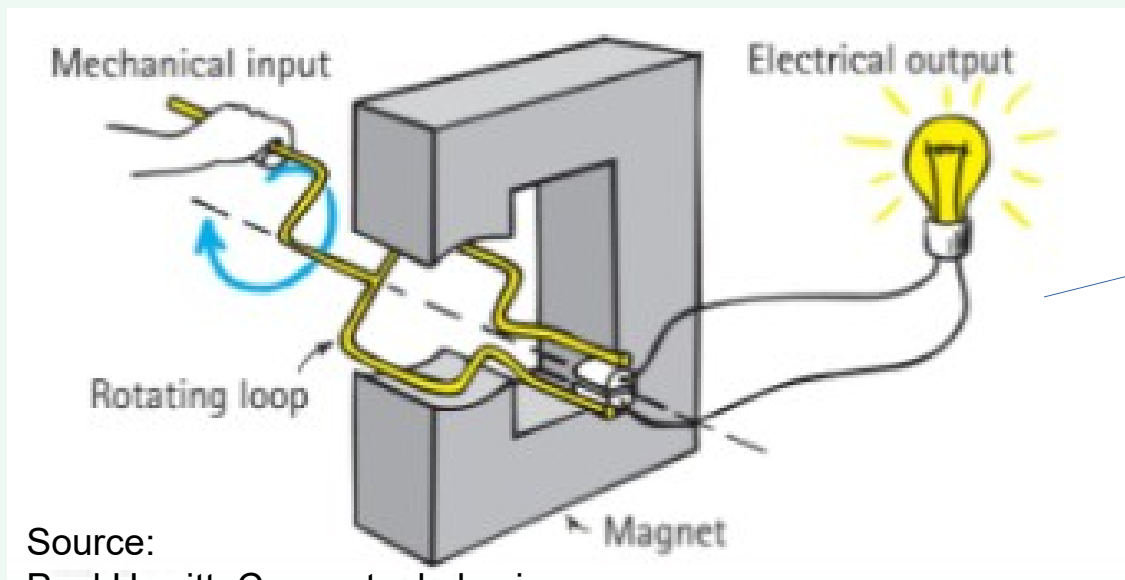


Electromagnetic Induction

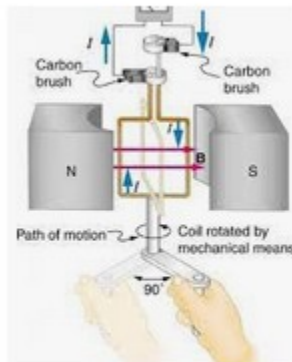


Source:
Paul Hewitt. Conceptual physics;
recommended reading

- historical notes
- EM induction, changing the magnetic flux through a close surface
- Faraday's law of induction, magnetic flux
- Lenz's law
- motional EMF
- applications and generator
- Some calculus, non-uniform field



Electric generator - Wikipedia



Electric Generators | Ph...

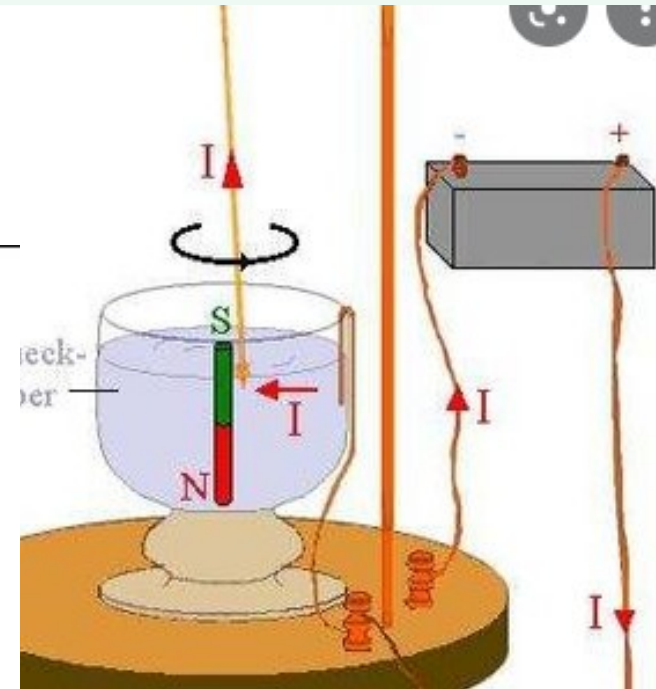
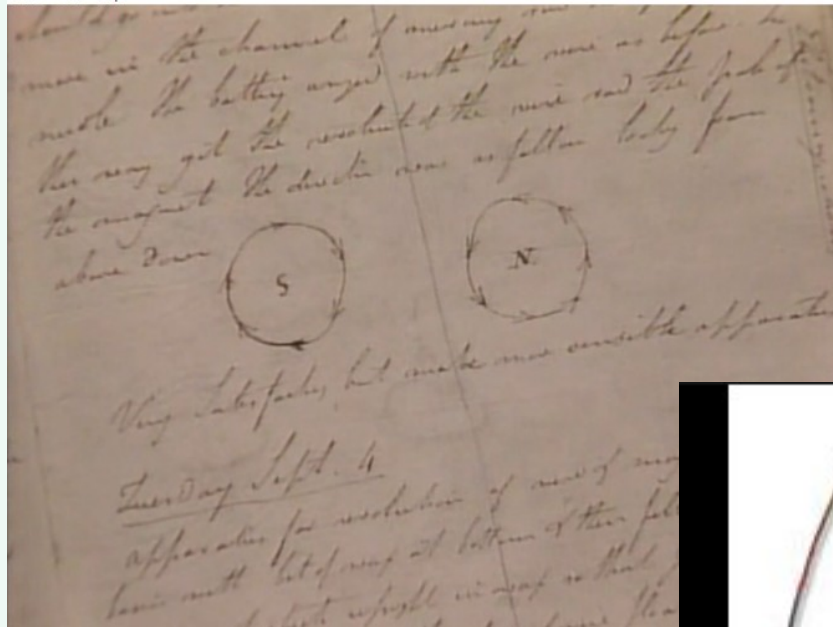


See animation – Changing flux $\rightarrow$ changing electricity induced.

It took 10 years for Michael Faraday to discover how to produce electricity with a magnetic field.

1821 (Oersted experiment) $\rightarrow$ 1831 Faraday's discovery.

He also built the first motor.



To generate electric current with B you need:

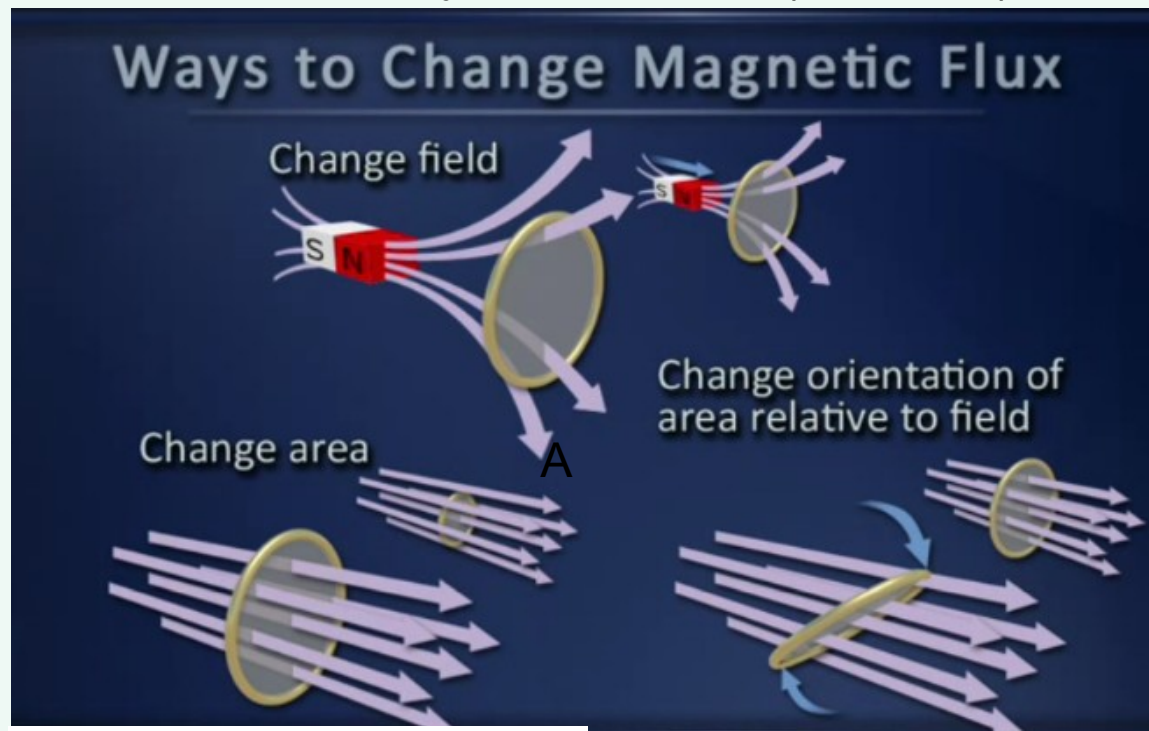
A conductor, a magnetic field and some relative motion conductor vs B

To generate our electricity today → conductor moves relative to B (field)

2 ways to do it

Faraday's law of induction→

Emf = change in the number of field lines poking through the area A defined by the conductor (loop here)



Faraday's Law of Induction

$$\mathcal{E} = -N \frac{\Delta \Phi_B}{\Delta t}$$

induced emf

number of loops

rate of change of flux with time

$$\Phi_B = |\vec{B}| \cos \theta |\vec{A}|$$

Flux.

Unit is the Weber (Wb)

Or T m^2 .

Magnetic field

$$= \Phi = B A$$

Area perpendicular to magnetic field B

Motional emf = Blv

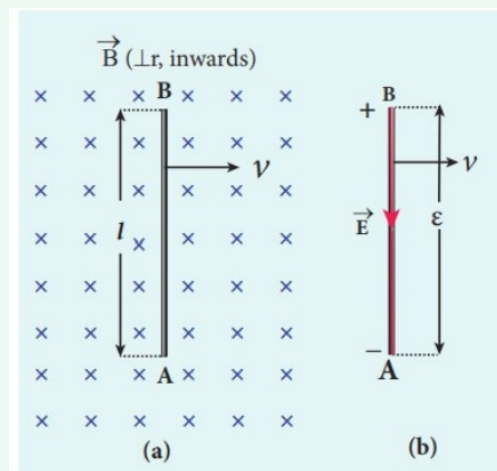
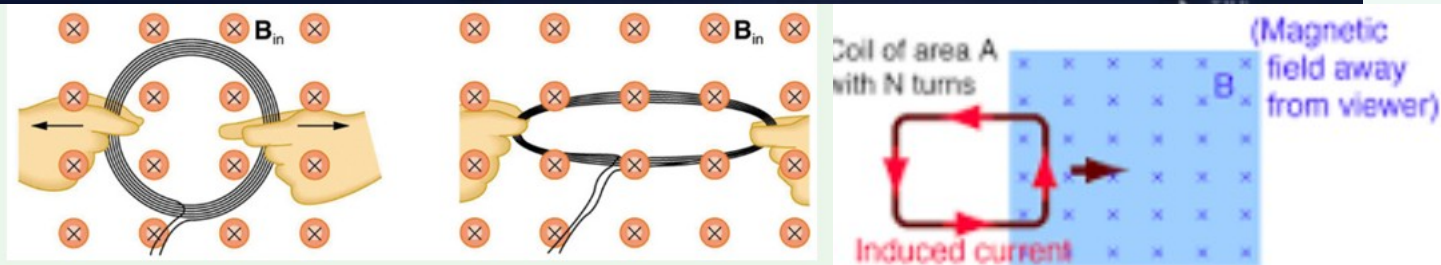
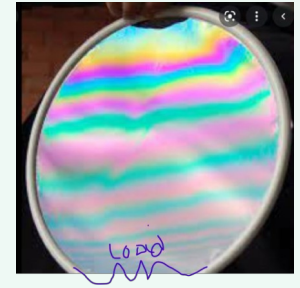
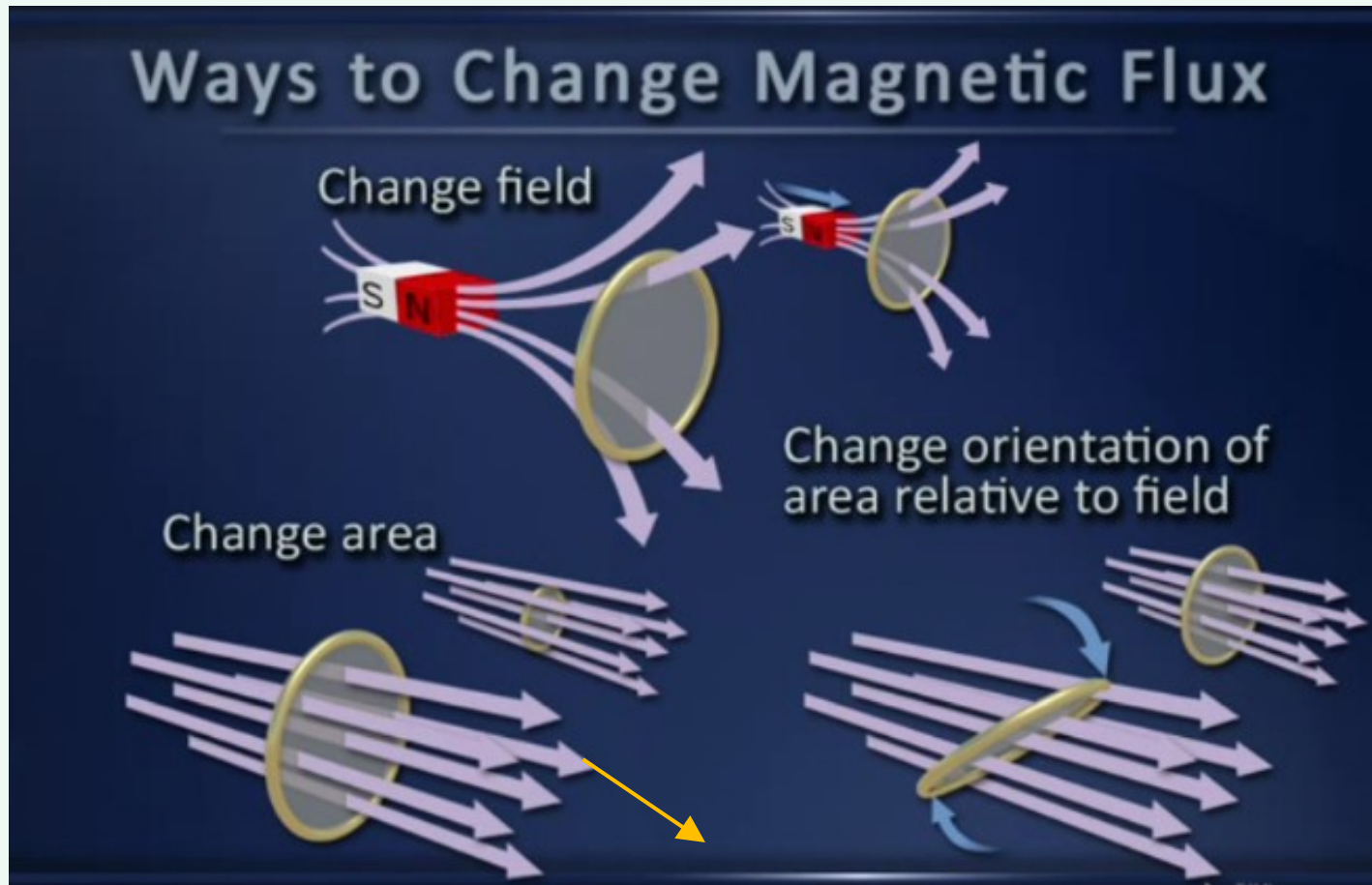


Figure 4.9 Motional emf from Lorentz force

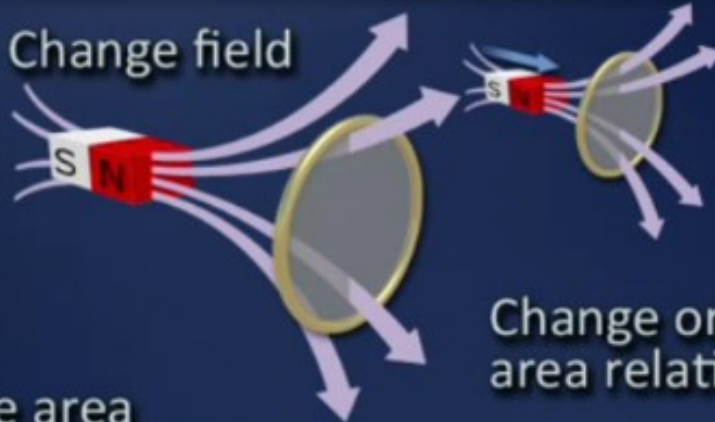
In those examples. The number of lines poking the “ghost area” attached to the Loop is changing with time. The emf is the rate at which the number of lines changes.



$$\text{emf induced} = N \times (\Delta\Phi/\Delta t)$$

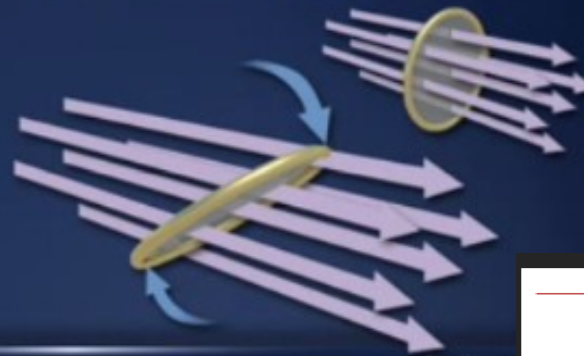
Ways to Change Magnetic Flux

Change field

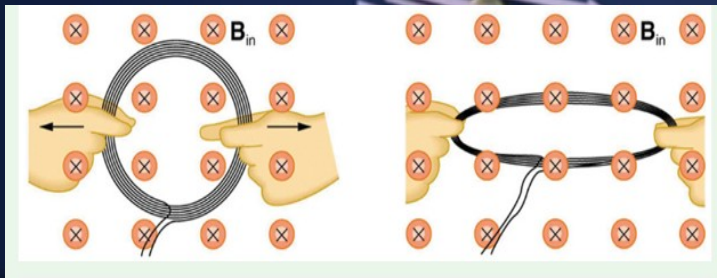


$$\text{emf induced} = N A (\Delta B / \Delta t)$$

Change orientation of area relative to field



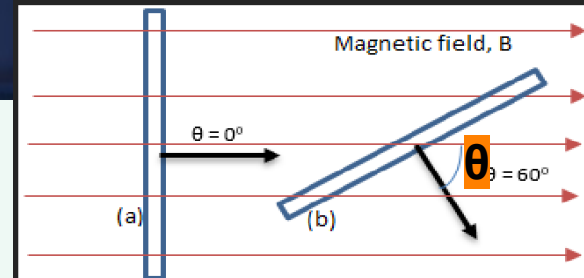
Change area



$$\text{emf induced} = N B \Delta A / \Delta t$$

$$\text{emf induced} = N B A \Delta(\cos(\theta)) / \Delta t$$

$$\text{emf induced} = B A \omega \sin(\omega t)$$



1. Black arrow is the direction of area placed in the magnetic field (red lines)
2. At (a) area vector and magnetic field have same direction; angle is zero and flux is maximum, 5 lines.
3. At (b) area vector makes 60 deg angle with the field. Flux has decreased, only 3 lines of B.

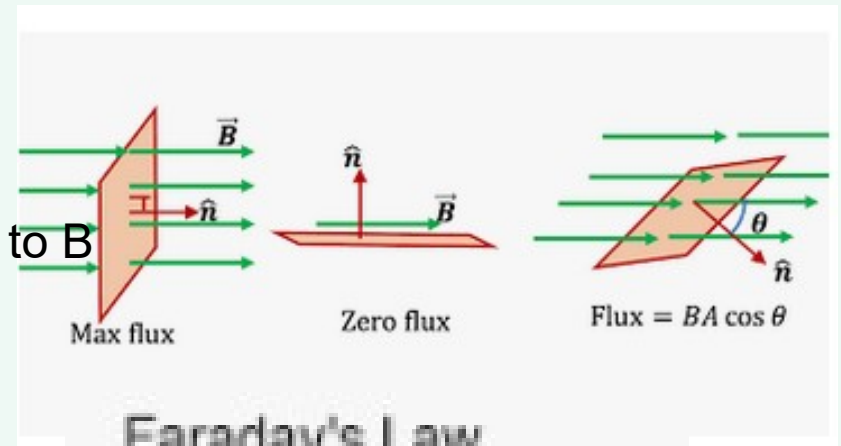
- Show demonstrations (don't miss class)
 - simulation gif
 - simulation html (106) and discuss flux
- And faradays law of induction.

Magnetic field

$$= \Phi = B A$$

Area perpendicular to magnetic field B

A cos(Θ) is the component perpendicular to B



Flux → $\Phi_B = B A \cos(\theta)$

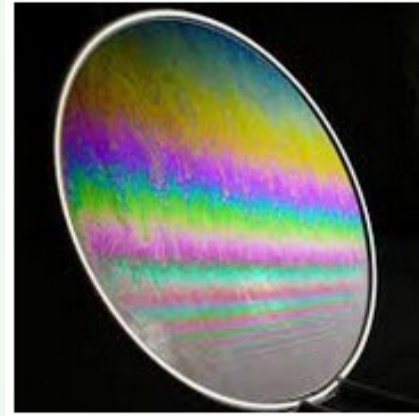
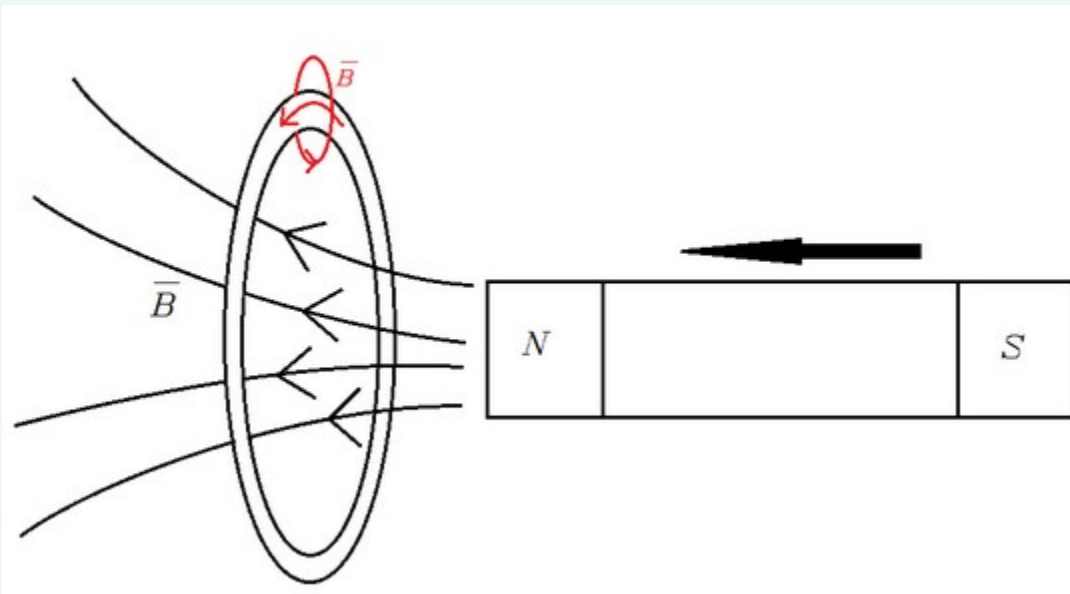
Greek letter Phi

B Field

Faraday's Law

$$\text{Emf} = -N \frac{\Delta \Phi}{\Delta t}$$

Lenz's Law



Can you see that the NUMBER OF LINES GOING THROUGH THE AREA Attached to the loop change with time (imaging a soap film with lines poking through)

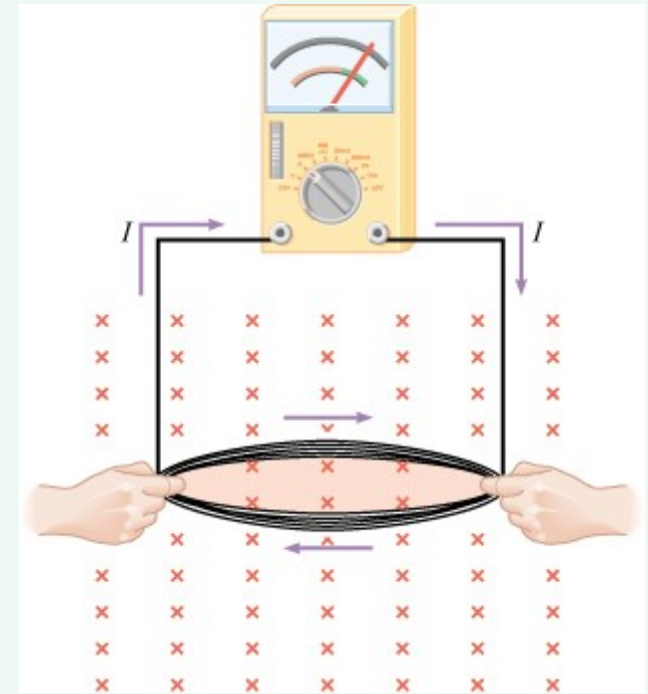
$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

The induced emf in a circuit is equal to the rate at which the magnetic flux through That circuit is changing with time. The emf is not a voltage drop or electric potential. It is the energy per unit charge gained as it moves around the loop. There is an induced electric field. Not the same as the one produced by static charge. The loop's rule (Kirchhoff rule) does not hold any more. (sum of voltage drops=0)

22.1 Induced Emf and Induced Current

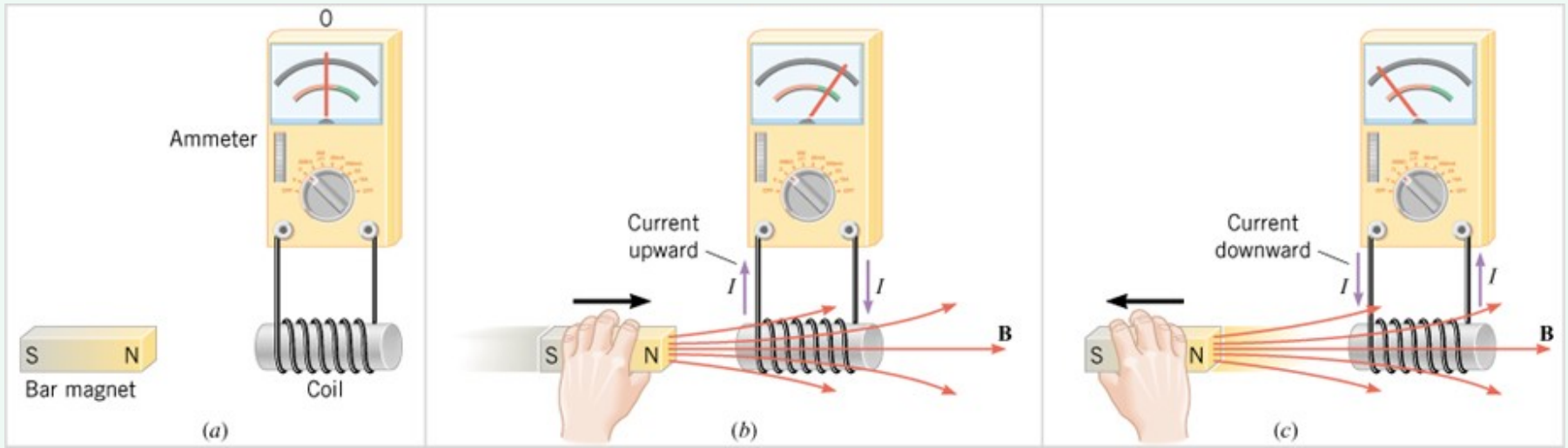
An emf can be induced by changing the area of a coil in a constant magnetic field

example1



In each example, both an emf and a current are induced because the coil is part of a complete circuit. If the circuit were open, there would be no induced current, but there would be an induced emf.

The phenomena of producing an induced emf with the aid of a magnetic field is called ***electromagnetic induction***.



It is the ***changing*** field that produces the current.

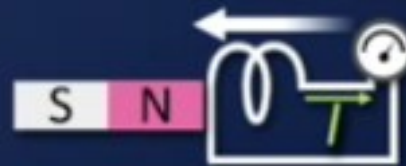
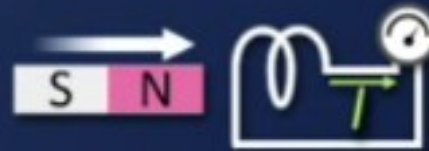
Changing magnetism $\rightarrow$ induced effect.

See how the number of lines poking through the cross section of the coil is Changing. What matters $\rightarrow$ numbers of magnetic field lines that change. Cut through the surface.

You can do the same with a loop of current = behaves like a magnet. You can also turn on and off a circuit (Faradays experiment). The magnetic field expands or collapses (see app).

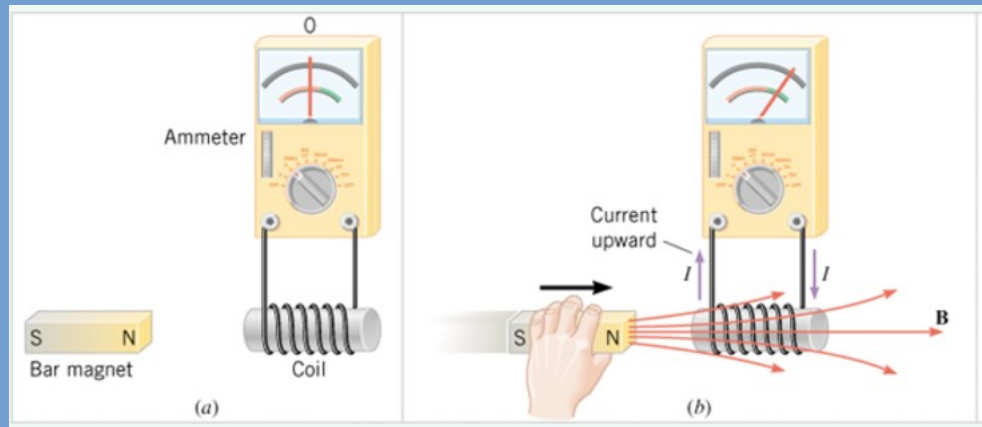
Induction Experiments

- Move magnet near coil:
induced current
- Move coil near magnet:
induced current
- Pass current through 1st coil;
move relative to 2nd coil:
induced current in 2nd coil
- Change current in 1st coil:
induced current in 2nd coil



Example 5 The Emf Induced by a Changing Magnetic Field

A coil of wire consists of 20 turns each of which has an area of 0.0015 m^2 . A magnetic field is perpendicular to the surface. Initially, the magnitude of the magnetic field is 0.050 T and 0.10s later, it has increased to 0.060 T . Find the average emf induced in the coil during this time.



Example 5 The Emf Induced by a Changing Magnetic Field

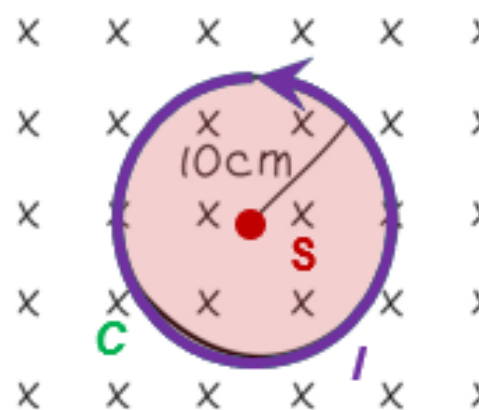
A coil of wire consists of 20 turns each of which has an area of 0.0015 m^2 . A magnetic field is perpendicular to the surface. Initially, the magnitude of the magnetic field is 0.050 T and 0.10 s later, it has increased to 0.060 T . Find the average emf induced in the coil during this time.

$$\begin{aligned} E &= -N \frac{\Delta \Phi}{\Delta t} = -N \frac{BA \cos \varphi - B_o A \cos \varphi}{\Delta t} \\ &= -NA \cos \varphi \left(\frac{B - B_o}{\Delta t} \right) = -(20)(0.0015 \text{ m}^2) \cos(0) \frac{0.060 \text{ T} - 0.050 \text{ T}}{0.10 \text{ s}} \\ &= -3.0 \times 10^{-3} \text{ V} \end{aligned}$$

A wire loop of radius 10 cm has resistance $2.0\ \Omega$.

The plane of the loop is perpendicular to a uniform $\mathbf{B}$ that's increasing at $0.10\ \text{T/s}$.

Find the magnitude of the induced current in the loop.

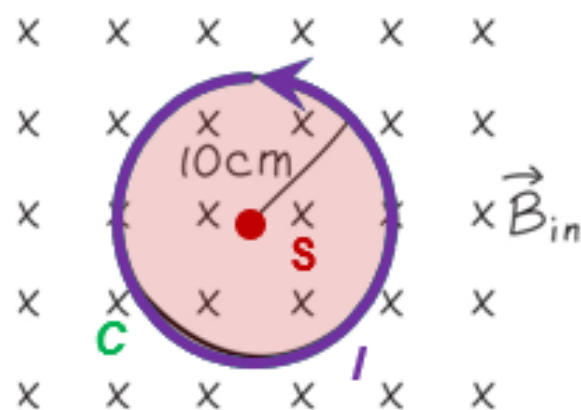


$\times 10^{-3}\ \text{V}$

A wire loop of radius 10 cm has resistance $2.0 \, \Omega$.

The plane of the loop is perpendicular to a uniform $\mathbf{B}$ that's increasing at $0.10 \, \text{T/s}$.

Find the magnitude of the induced current in the loop.



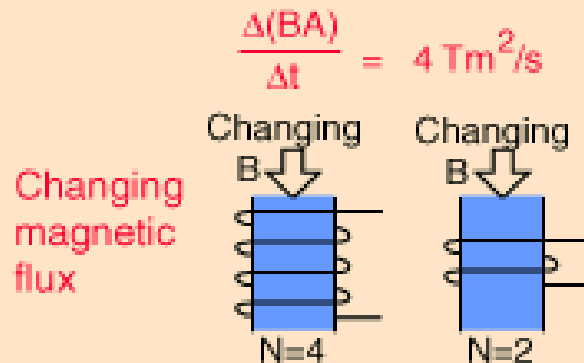
$$\Phi_B = -BA = -B \pi r^2$$

$$\frac{d\Phi_B}{dt} = \frac{d}{dt}(-B \pi r^2) = -\pi r^2 \frac{dB}{dt}$$

$$E = -\frac{d\Phi_B}{dt} = \pi (0.1 \, \text{m})^2 (0.10 \, \text{T/s}) = 3.14 \times 10^{-3} \, \text{V}$$

$$I = \frac{E}{R} = \frac{3.14 \times 10^{-3} \, \text{V}}{2.0 \, \Omega} = 1.57 \times 10^{-3} \, \text{A}$$

CCW

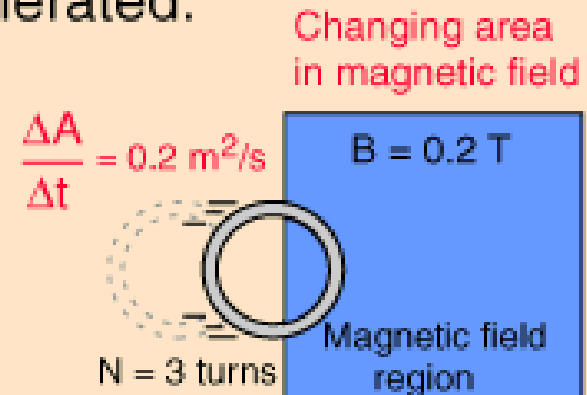


Case 1

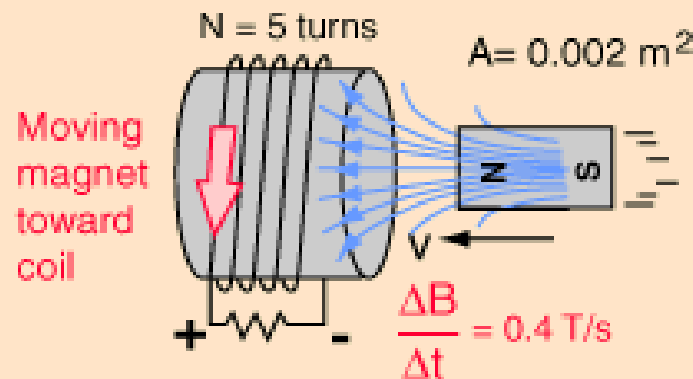
Voltage generated $= -N \frac{\Delta(BA)}{\Delta t}$

Faraday's Law

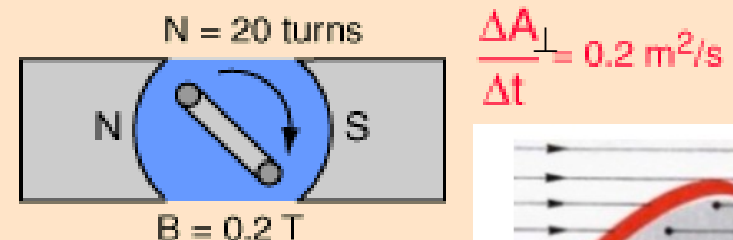
Faraday's Law summarizes the ways voltage can be generated.



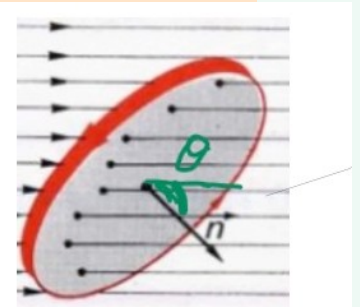
Case 2



Case 3

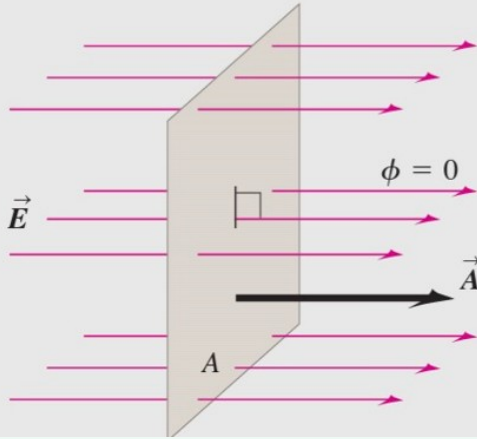


Case 4 : rotating coil in B
 Here $A_{\perp} = A \cos(\theta)$
 See next slide



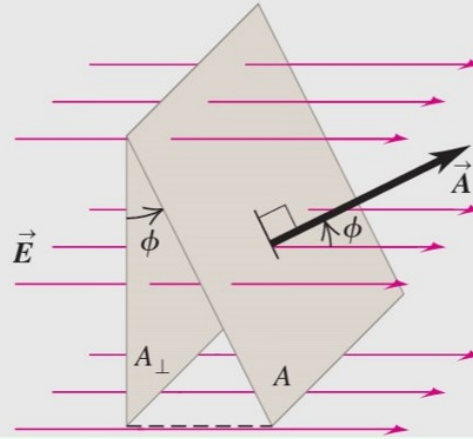
(a) Surface is face-on to electric field:

- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.



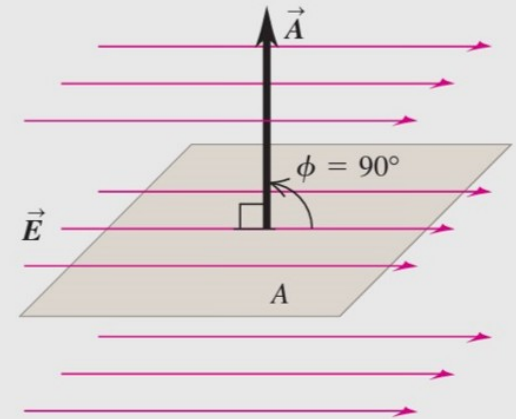
(b) Surface is tilted from a face-on orientation by an angle ϕ :

- The angle between $\vec{E}$ and $\vec{A}$ is ϕ .
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.



(c) Surface is edge-on to electric field:

- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.



$$A_{\perp} = A \cos(\theta)$$

$$\Phi = N \vec{A} \cdot \vec{B} = AB \cos(\theta) = A_{\perp} B$$

Rotation of a coil in uniform $\vec{B} \rightarrow A_{\perp}$ changes
 θ changes

Let there be light!

Lamp requires 3 V to
light at full brightness

Connect in a wire loop
20 cm in diameter

At what rate should the
magnetic field change?



Let there be light!

Rate of change: $\frac{\Delta\Phi}{\Delta t} = \pi R^2 \frac{\Delta B}{\Delta t}$

Faraday's law:
Induced voltage is rate
of change of flux

So: want $\Delta\Phi/\Delta t = 3 \text{ V}$

Solve for $\Delta B/\Delta t$:

$$\frac{\Delta B}{\Delta t} = \left(\frac{1}{\pi R^2} \right) \frac{\Delta\Phi}{\Delta t} = \left(\frac{1}{\pi (0.1 \text{ m})^2} \right) (3 \text{ V}) = 95 \text{ T/s}$$



2

A conducting coil with a radius of 2.5cm has 150 turns. The coil moves perpendicularly to a magnetic field that becomes stronger at the rate of 1.8 mT/s. Find the magnitude of the electromotive force induced in the coil. Give your answer in millivolts to two decimal places.

Faraday's Law

3

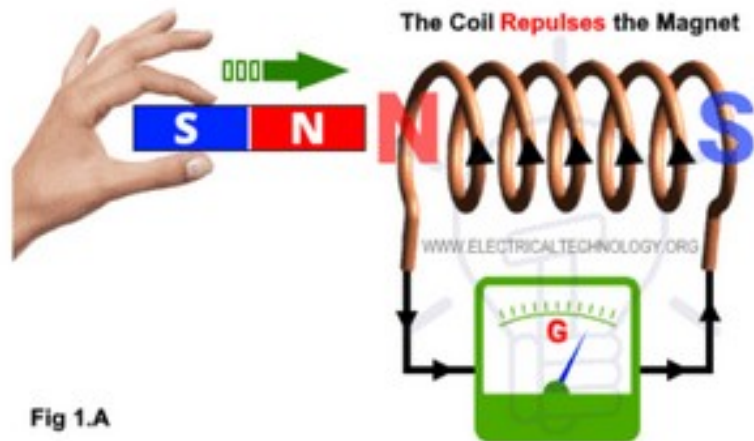
A constant uniform magnetic field of 0.50 T is applied at right angles to the plane of a flat rectangular loop of area $3.0 \times 10^{-3} \text{ m}^2$. If the area of this loop changes steadily from its original value to a new value of $1.6 \times 10^{-3} \text{ m}^2$ in 1.6 s, what is the emf induced in the loop?

4

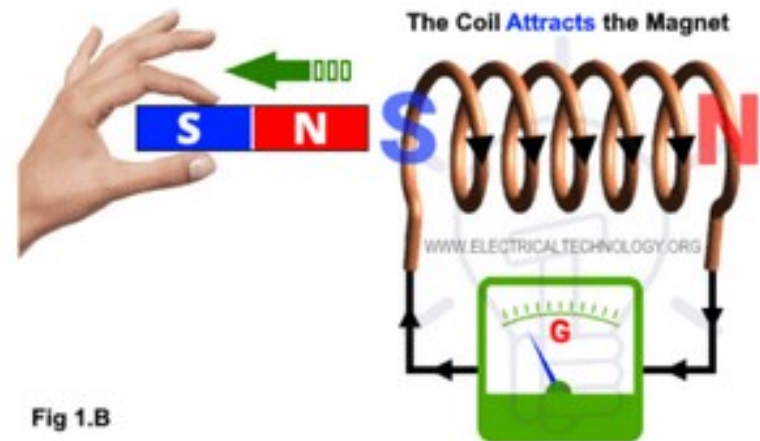
A round flat metal coil has 180 turns and negligible resistance. It is connected in a series circuit with a $17\text{-}\Omega$ resistor, with nothing else in the circuit. You measure that a 6.0-A current flows through the resistor when a magnetic field through the coil, perpendicular to its area, is changing at 3.0 T/s. What is the radius of the coil?

LENZ'S LAW

An induced Current always flows in a direction such that it opposes the change which produced it.



When the "N" Pole of the magnet is moved towards the coil, end of the coil becomes "N" Pole.



When the "N" Poles of the magnet is moved away from the coil, end of the coil becomes "S" Pole.

Conservation of energy

Lenz is so important; it has its own name.
-> conservation of energy. Opposition to the change of flux.

Lenz's Law

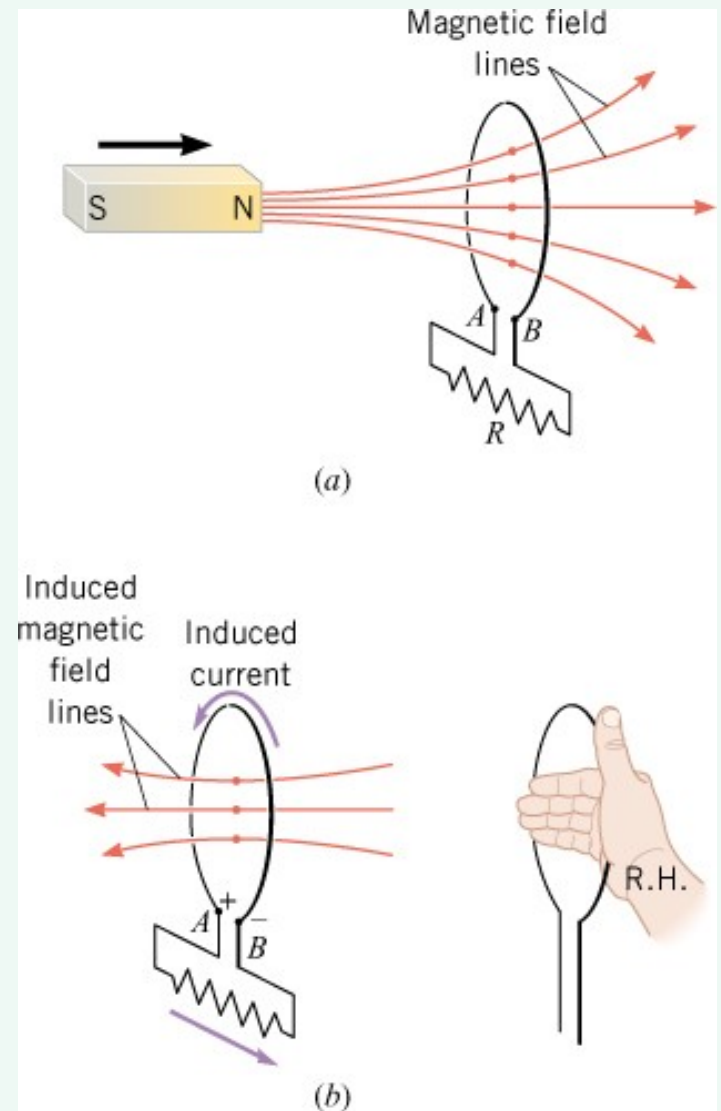
- The direction of an induced voltage or current is such as to oppose the change giving rise to the induced effect
- Essential for consistency with energy conservation!
- Mathematically: minus sign in Faraday's law

$$V_{\text{induced}} = - \frac{\Delta \Phi_B}{\Delta t}$$

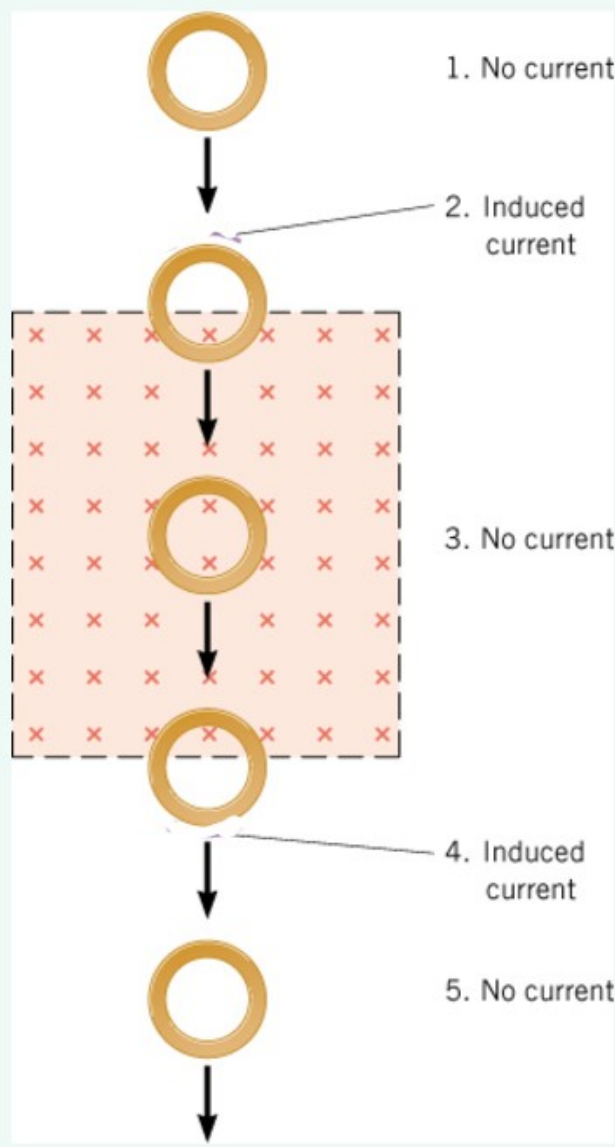
Video lenz and simulation

***Conceptual Example 8* The Emf Produced by a Moving Magnet**

A permanent magnet is approaching a loop of wire. The external circuit consists of a resistance. Find the direction of the induced current and the polarity of the induced emf.



22.5 Lenz's Law



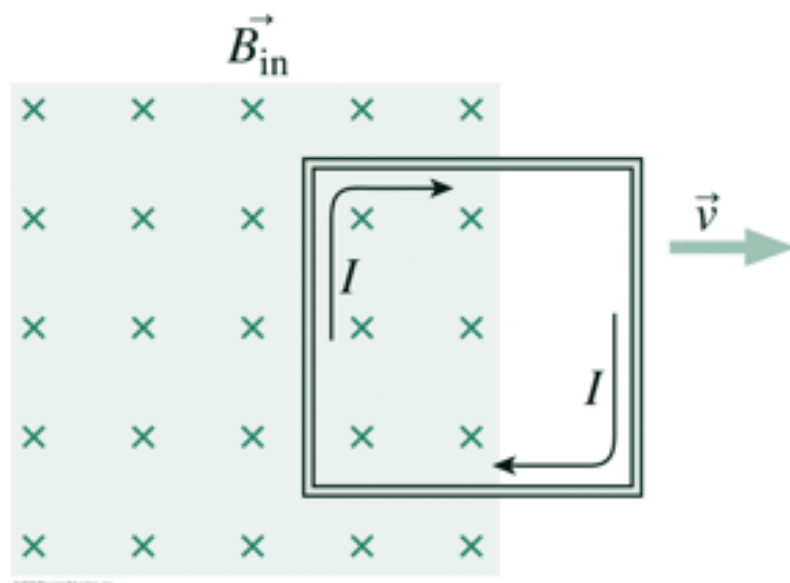
Conceptual Example 9 The Emf Produced by a Moving Copper Ring.

There is a constant magnetic field directed into the page in the shaded region. The field is zero outside the shaded region. A copper ring slides through the region.

For each of the five positions, determine whether an induced current exists and, if so, find its direction ?

Use Lenz law

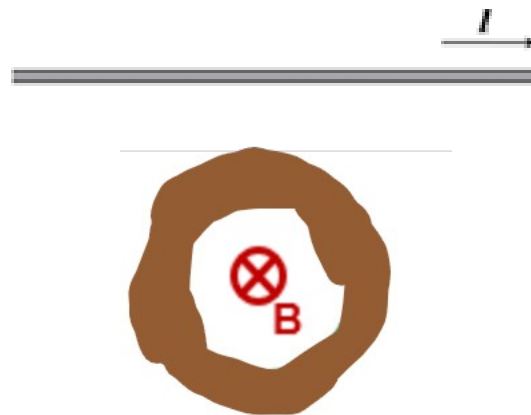
- What will be the direction of the current in the loop when it first enters the field shown, coming into the field from the left side?
- A. Clockwise
- B. Counterclockwise



A long wire carries a current I as shown.

What's the direction of the current in the circular conducting loop when I is

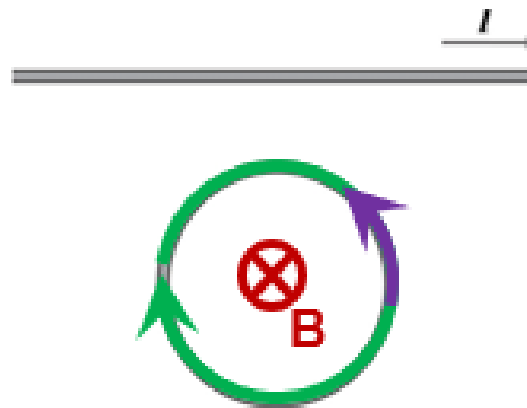
- (a) increasing and
- (b) decreasing?



A long wire carries a current I as shown.

What's the direction of the current in the circular conducting loop when I is

- CCW
CW
- (a) increasing and
(b) decreasing?



Setting $\mathbf{n} \parallel \mathbf{B}$

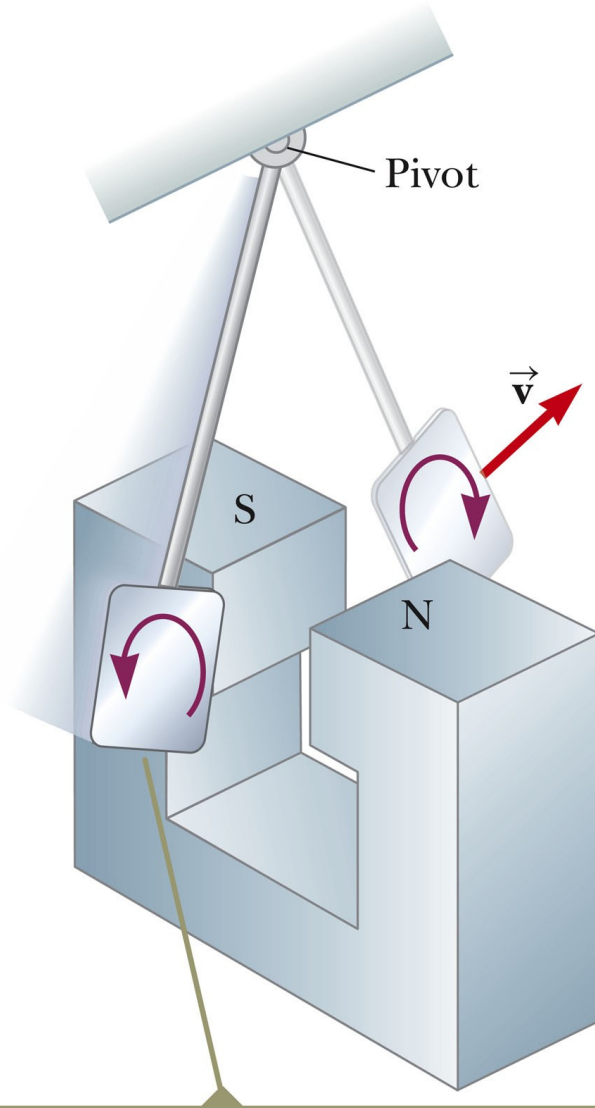
$\rightarrow$ C is CW & $\Phi > 0$

$I \uparrow \rightarrow B \uparrow$

$\rightarrow d\Phi/dt > 0$

$\rightarrow E < 0$

$\therefore I_{\text{ind}}$ is CCW



As the plate enters or leaves the field, the changing magnetic flux induces an emf, which causes eddy currents in the plate.

The changing magnetic field creates eddy currents and the plate stops. That's how magnetic brakes work.

See video

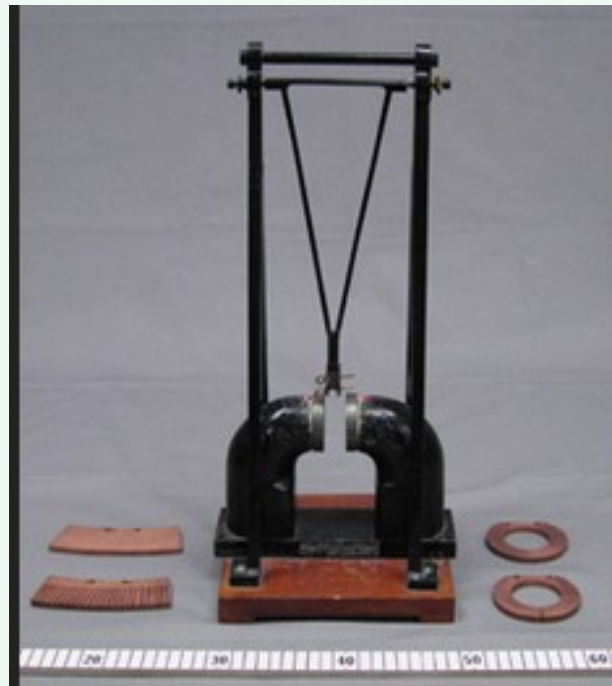
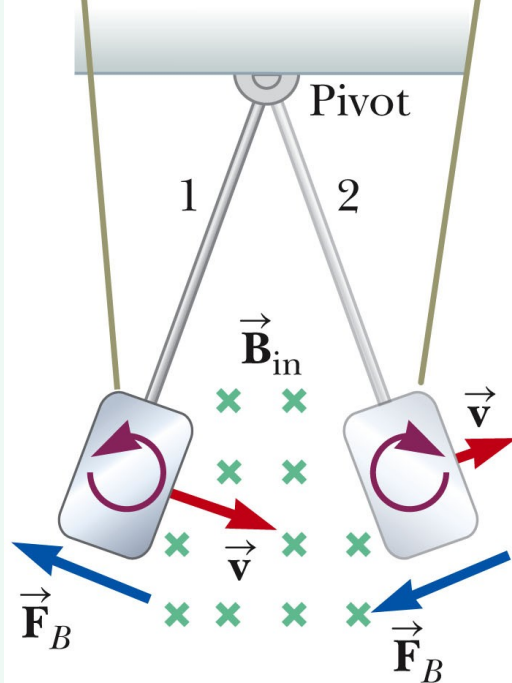


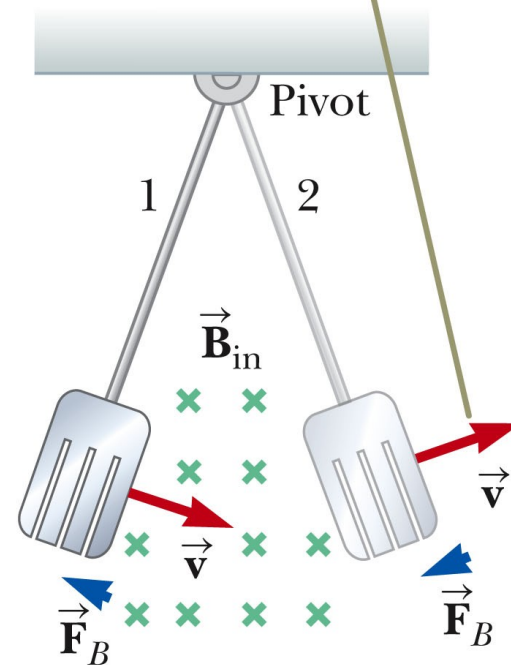
Fig. 31-21, p. 911

As the conducting plate enters the field, the eddy currents are counterclockwise.

As the plate leaves the field, the currents are clockwise.

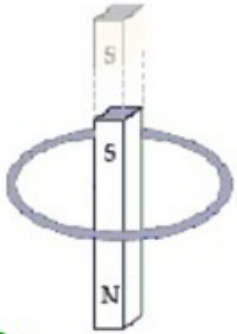


When slots are cut in the conducting plate, the eddy currents are reduced and the plate swings more freely through the magnetic field.



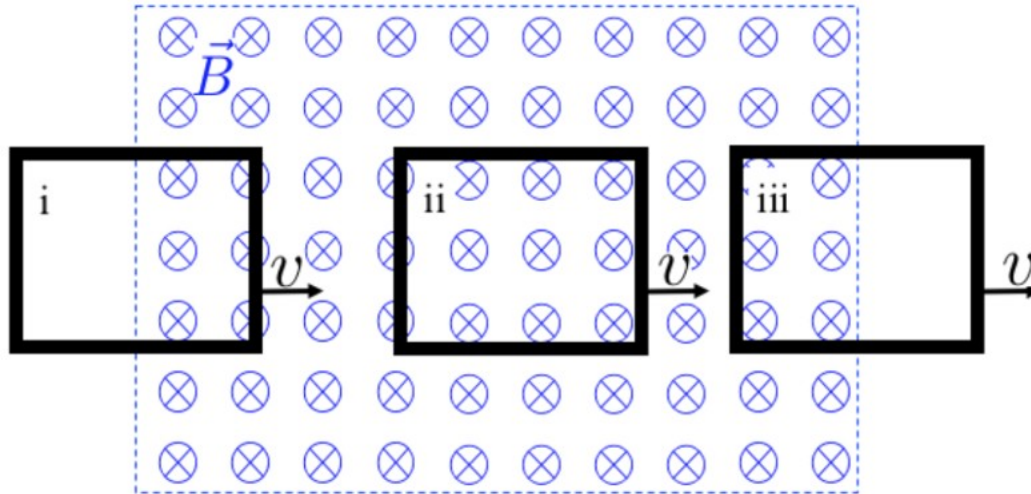
<https://www.youtube.com/watch?v=N7tli71-AjA> demo magnet/aluminum falling in copper pipe**b**

A bar magnet is dropped from above and falls through the loop of wire shown below. The north pole of the bar magnet points downward towards the page as it falls. Which statement is correct?



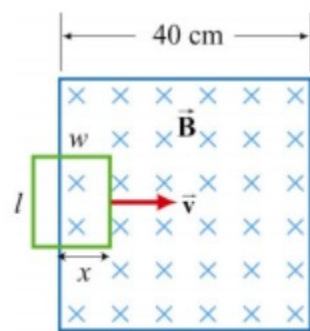
Lenz law. First the current flows to repel the magnet then to attract it.

Motional emf → converting mechanical energy to electrical energy



A small rectangular loop of length $l = 10$ cm and width $w = 8.0$ cm with resistance $R = 2.0 \Omega$ is pulled at a constant speed $v = 2.0$ cm/s through a region of uniform magnetic field $B = 2.0$ T, pointing into the page, as shown in Figure 10.11.6.

1



A) Find the emf. Hint: its the area That is changing. The area is $A = x l$ (through which B is poking through). And $\Delta x / \Delta t = \text{speed}$

B) find the induced current

Figure 10.11.6

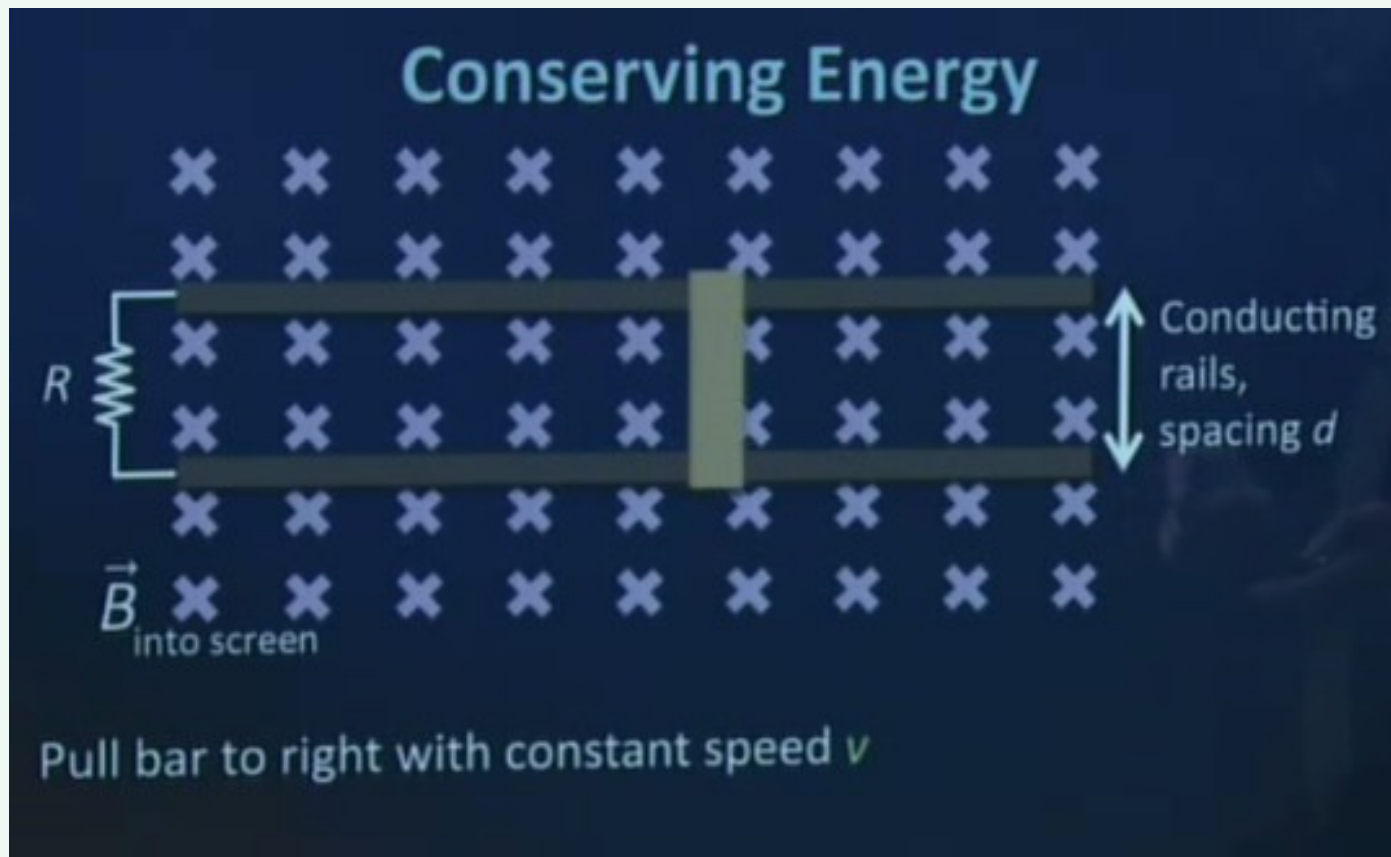
Conservation of energy, lenz law

The bar is pulled to the right speed is v (constant). The width is L . B is constant.

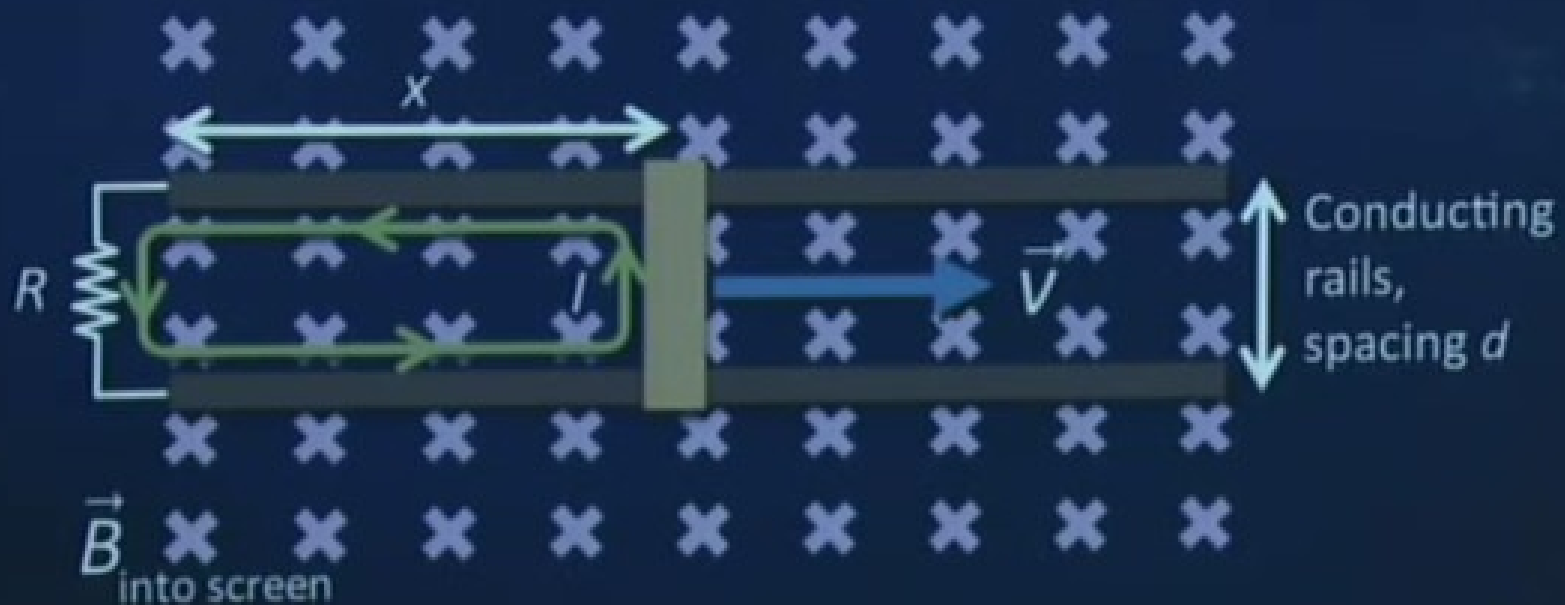
1- Area increases $\rightarrow$ emf is induced, current is induced. Find the induced emf, induced current, the electric power dissipated through R / (as a function of B, L, v, R)

2- you pull on the bar with a force F_u at a speed v . But there is another force acting on the bar. $F_b = ILB$ that is opposing your force. Speed is constant.

What is the work done by you?



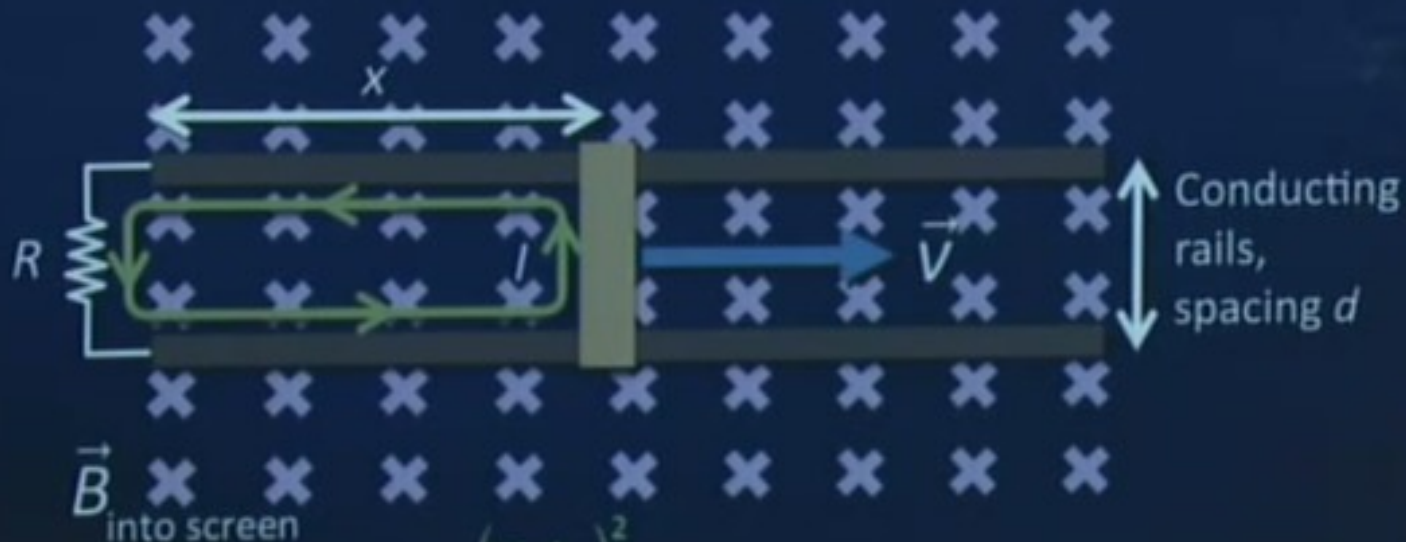
Conserving Energy



$$\frac{\Delta \Phi}{\Delta t} = Bdv \quad \text{Faraday's law: } |V| = \frac{\Delta \Phi}{\Delta t} = Bdv$$

$$\text{Ohm's law: } I = \frac{V}{R} = \frac{Bdv}{R} \quad \text{Direction: counterclockwise}$$

Conserving Energy



Electrical power: $P = \frac{(Bdv)^2}{R}$

Force you need to apply for constant velocity: $F = \frac{B^2 d^2 v}{R}$ to right

Rate at which you do work: $P = Fv = \frac{(Bdv)^2}{R}$

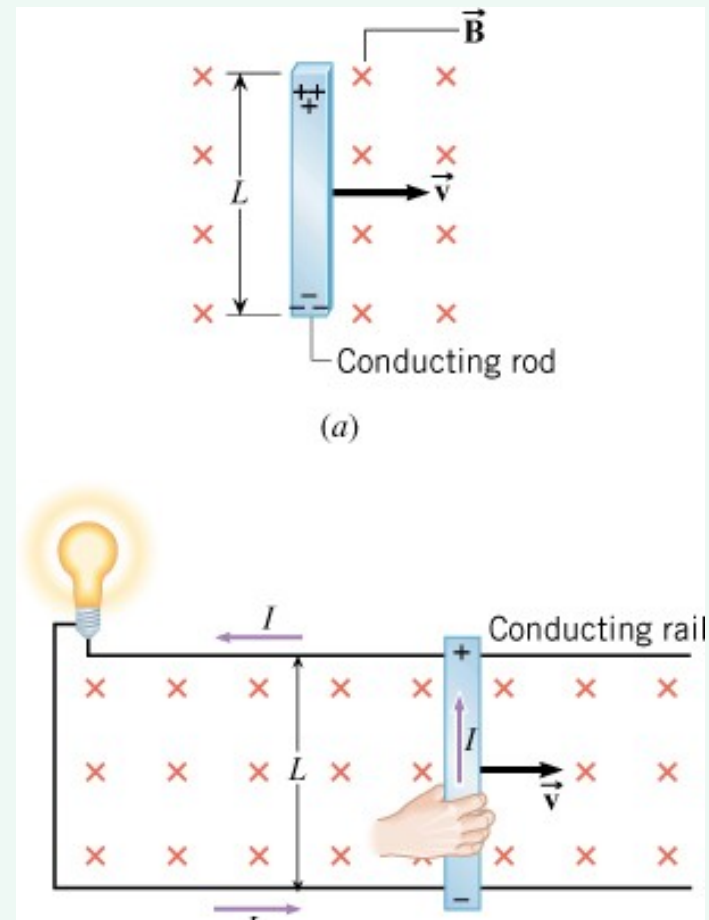
- If you lower the electrical resistance connected across a generator while turning the generator at constant speed, how will turning the generator change?
 - A. It will get harder to turn the generator.
 - B. It will get easier to turn the generator.
 - C. You will not notice any change in turning the generator.

Hint": use previous result. $P = B^2 d^2 v^2 / R$

22.2 Motional Emf

Example 1 Operating a Light Bulb with Motional Emf

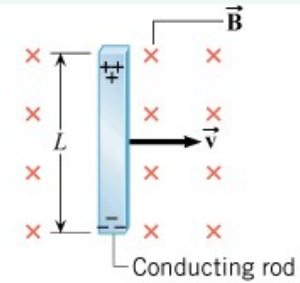
Suppose the rod is moving with a speed of 5.0 m/s perpendicular to a 0.80-T magnetic field. The rod has a length of 1.6 m and a negligible electrical resistance. The rails also have a negligible electrical resistance. The light bulb has a resistance of 96 ohms . Find (a) the emf produced by the rod and (b) the current induced in the circuit.



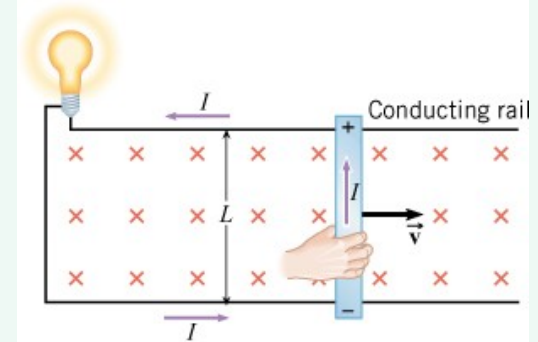
22.2 Motional Emf

(a) $E = vBL = (5.0 \text{ m/s})(0.80 \text{ T})(1.6 \text{ m}) = 6.4 \text{ V}$

(b) $I = \frac{E}{R} = \frac{6.4 \text{ V}}{96 \Omega} = 0.067 \text{ A}$



(a)



(b)

The magnetic field is constant. In the screen. The bar is mobile. The power supply provides V_b and a current I flows. There is a $F_m = I d B$ acting on the bar. The bar starts to move but there is an emf opposing the motion because it does not want the flux to increase. If R is negligible: $V_b = \text{emf}$
 $V_b = Blv$ so terminal speed = V_b/I it works like a motor if no load.

If R is not negligible: $V_b - \text{emf} = R I_{\text{total}}$ – the speed is lower

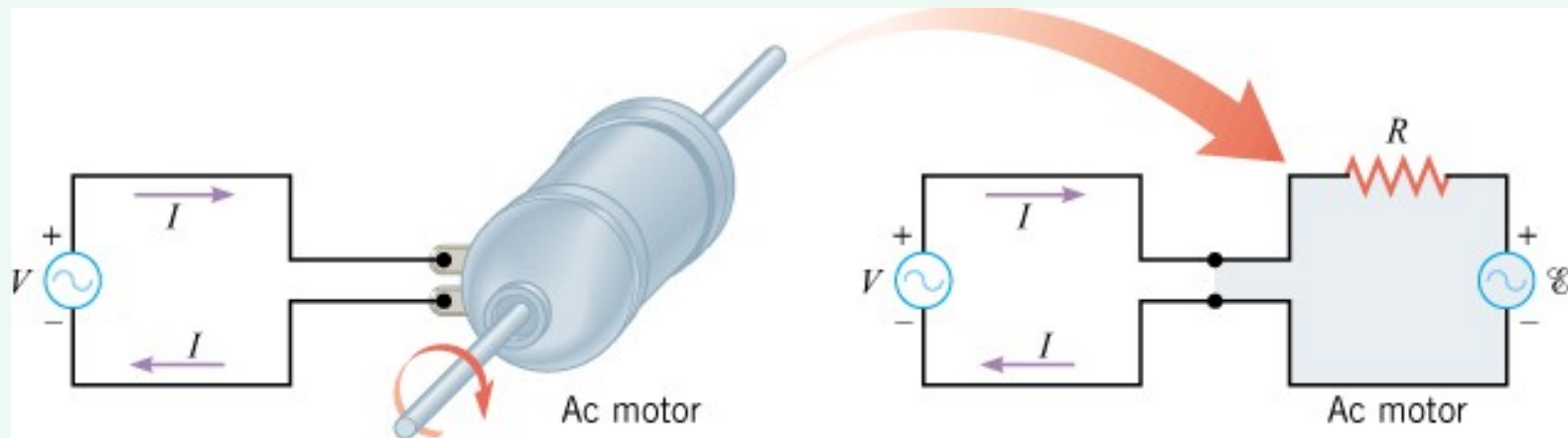
In the example “Conserving Energy,” replace the resistor with a battery and leave the bar initially at rest



- Describe qualitatively the bar's subsequent motion
- Find an expression for its final speed

THE BACK EMF GENERATED BY AN ELECTRIC MOTOR

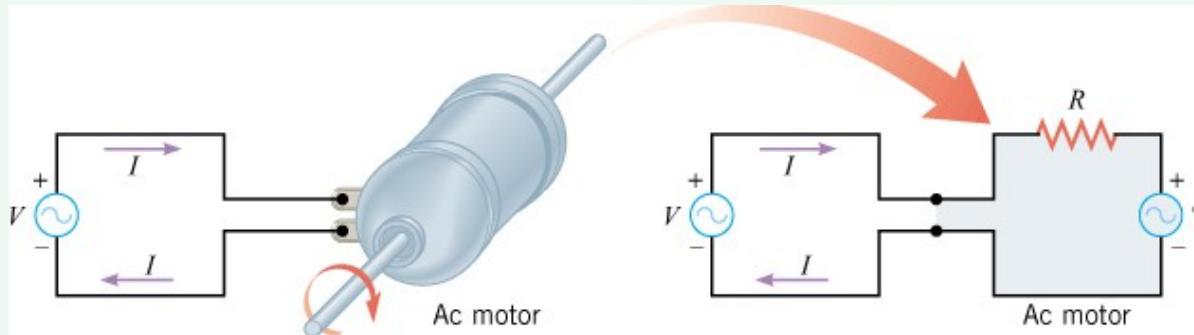
When a motor is operating, two sources of emf are present: (1) the applied emf V that provides current to drive the motor, and (2) the emf induced by the generator-like action of the rotating coil.



In the slide 33 example, electric energy is converted to mechanical energy. The emf (or back emf or induced voltage) opposes the applied voltage. And the bar moves at a constant speed. What happens in a motor? If there is no load, emf opposes V_{applied} and $V_{\text{applied}} = \text{emf}$ after a while like previously. But if there is a load, the load slows down the motor $\rightarrow$ emf decreases $\rightarrow$ more current is available to provide the torque. There is an equilibrium $I = (V_{\text{applied}} - \text{emf})/R$. Emf helps in the conversion of energy.

22.7 The Electric Generator

Consistent with Lenz's law, the induced emf acts to oppose the applied emf and is called **back emf**.



$$I = \frac{V - E}{R}$$

<https://circuitglobe.com/what-is-back-emf-in-dc-motor.html>

Example 12 Operating a Motor

The coil of an ac motor has a resistance of 4.1 ohms. The motor is plugged into an outlet where the voltage is 120.0 volts (rms), and the coil develops a back emf of 118.0 volts (rms) when rotating at normal speed. The motor is turning a wheel. Find (a) the current when the motor first starts up and (b) the current when the motor is operating at normal speed.

Back emf is the generator output of a motor, and so it is proportional to the motor's angular velocity ω . It is zero when the motor is first turned on, meaning that **the coil receives the full driving voltage** and the motor draws maximum current when it is on but not turning.

Example 12 Operating a Motor

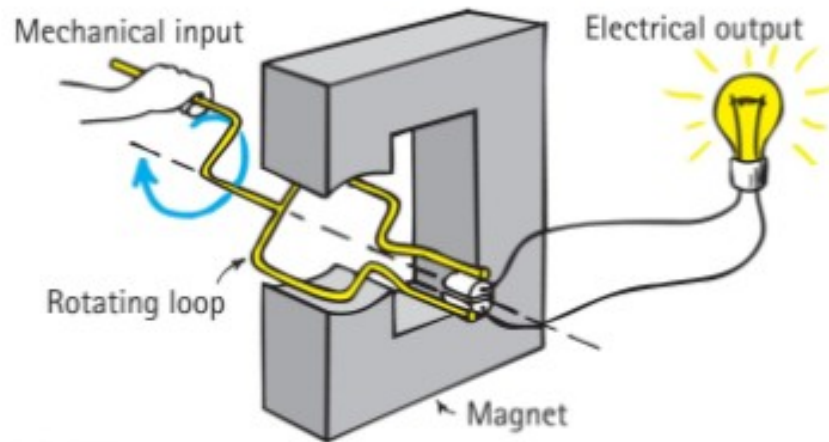
The coil of an ac motor has a resistance of 4.1 ohms. The motor is plugged into an outlet where the voltage is 120.0 volts (rms), and the coil develops a back emf of 118.0 volts (rms) when rotating at normal speed. The motor is turning a wheel. Find (a) the current when the motor first starts up and (b) the current when the motor is operating at normal speed.

$$(a) \quad I = \frac{V - E}{R} = \frac{120 \text{ V} - 0 \text{ V}}{4.1 \Omega} = 29 \text{ A}$$

$$(b) \quad I = \frac{V - E}{R} = \frac{120 \text{ V} - 118.0 \text{ V}}{4.1 \Omega} = 0.49 \text{ A}$$

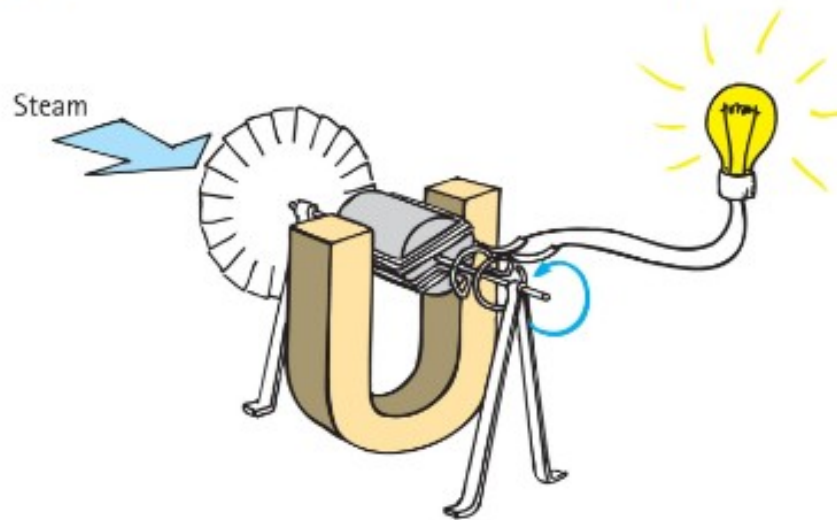
Generators and Alternating Current

- Generator
 - Opposite of a motor
 - Converts mechanical energy into electrical energy via coil motion
 - Produces alternating voltage and current



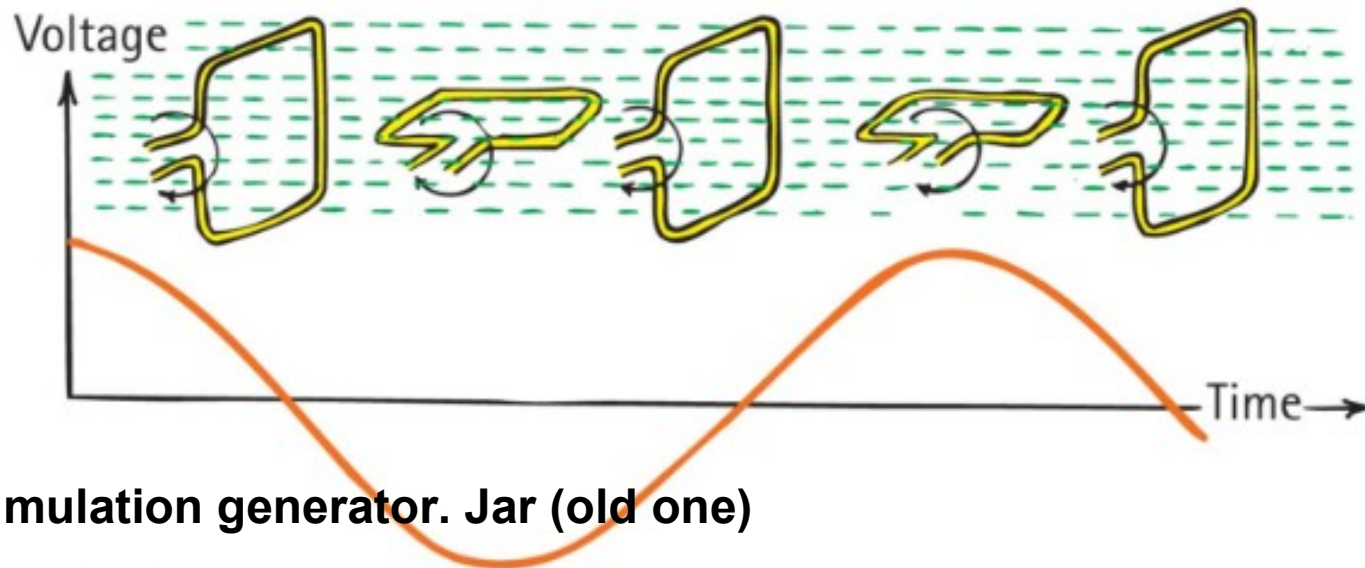
Power Production

- Using Faraday and Henry's discovery of electromagnetic induction, Nikola Tesla and George Westinghouse showed that electricity could be generated in sufficient quantities to light cities.



Generators and Alternating Current, Continued

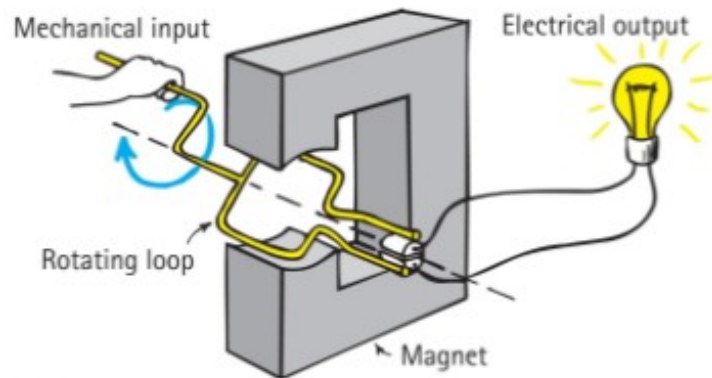
- The frequency of alternating voltage induced in a loop is equal to the frequency of the changing magnetic field within the loop.



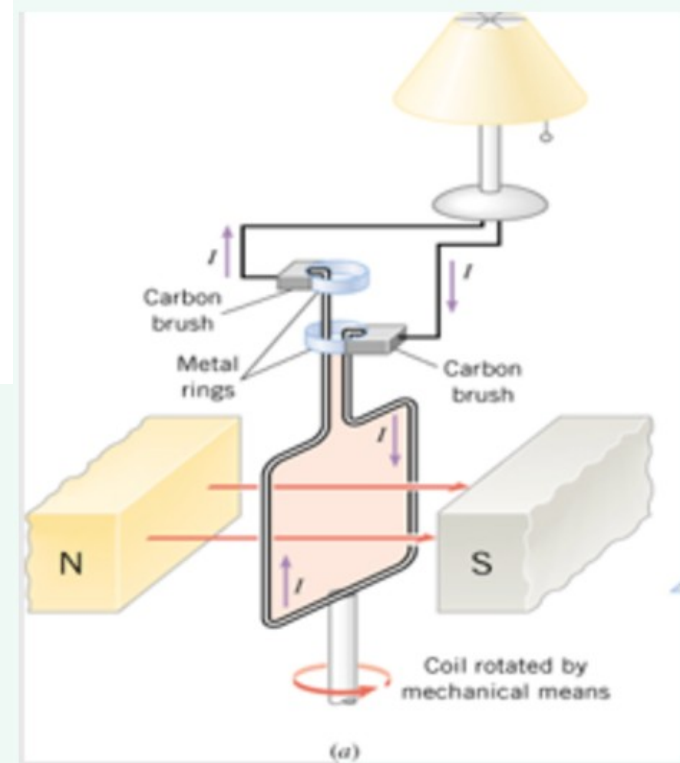
Show simulation generator. Jar (old one)

Generators and Alternating Current

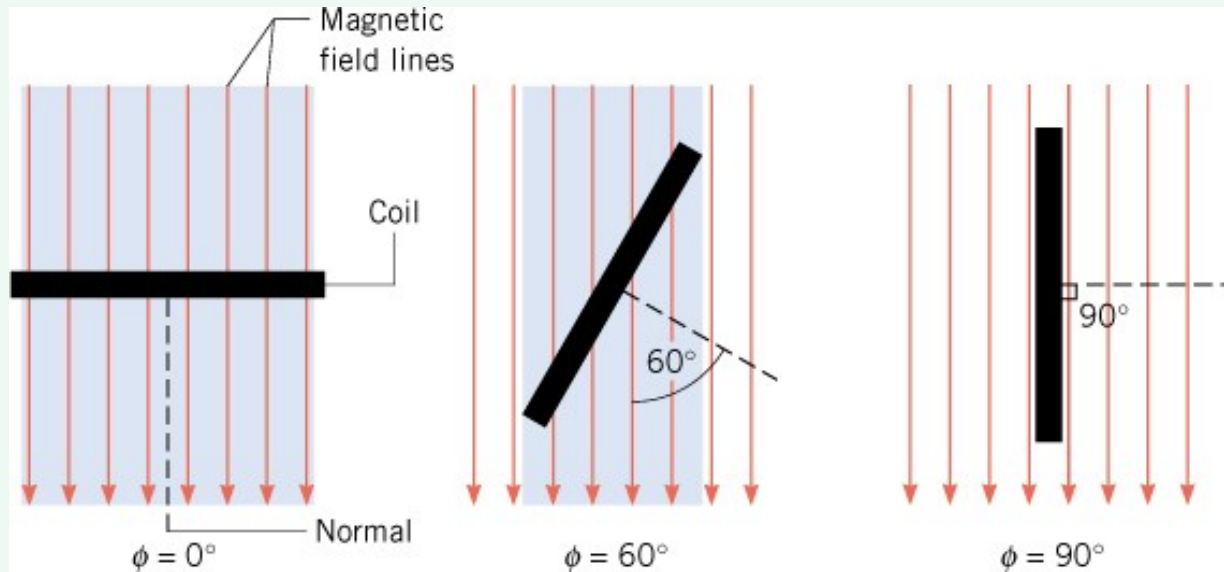
- Generator
 - Opposite of a motor
 - Converts mechanical energy into electrical energy via coil motion
 - Produces alternating voltage and current



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Electric generator design



$$\begin{aligned}\phi &= NBA \cos \theta \\ \theta &= \omega t \quad \left(\omega = \frac{2\pi}{T} \right) \\ \frac{d\phi}{dt} &= \mathcal{E}_{\text{mf}} = \frac{\omega NBA \sin(\omega t)}{\mathcal{E}_{\text{mf}} \text{max}}\end{aligned}$$

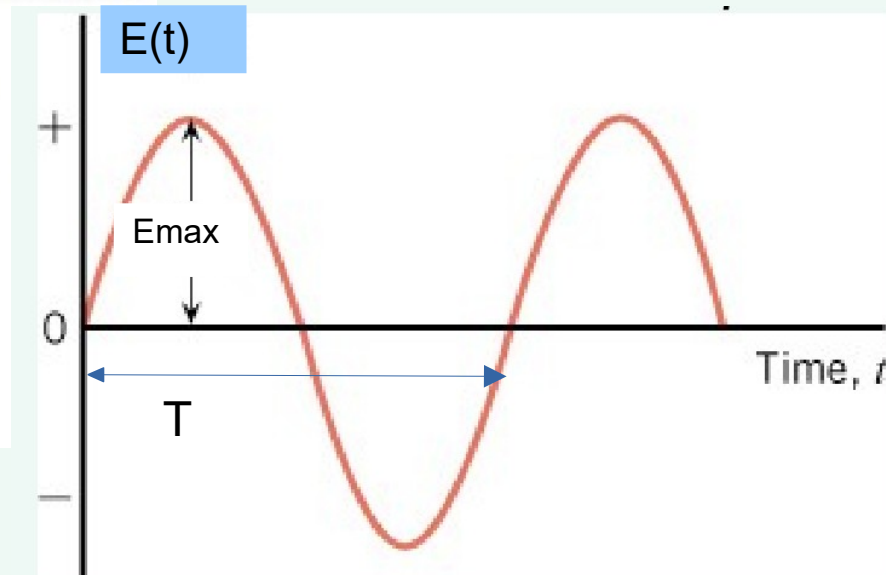
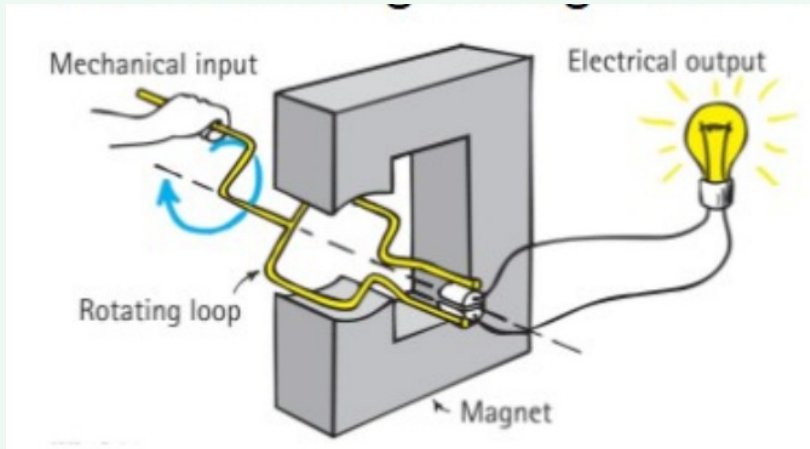
Video AC current

Spin a coil in a magnetic field → get alternating current

An **AC generator** (alternating current generator) is a device that converts mechanical energy

(like rotation) into electrical energy in the form of alternating current (AC), where the current periodically reverses direction.

The induced electromotive force (EMF) in an AC generator is alternating (AC)



$$E_{max} = N A B \omega$$

$$\omega = 2\pi/T = 2\pi f$$

T is the period of the sine wave. f the frequency $= 1/T$

See next slide

$$E(t) = N A B \omega \sin(\omega t)$$

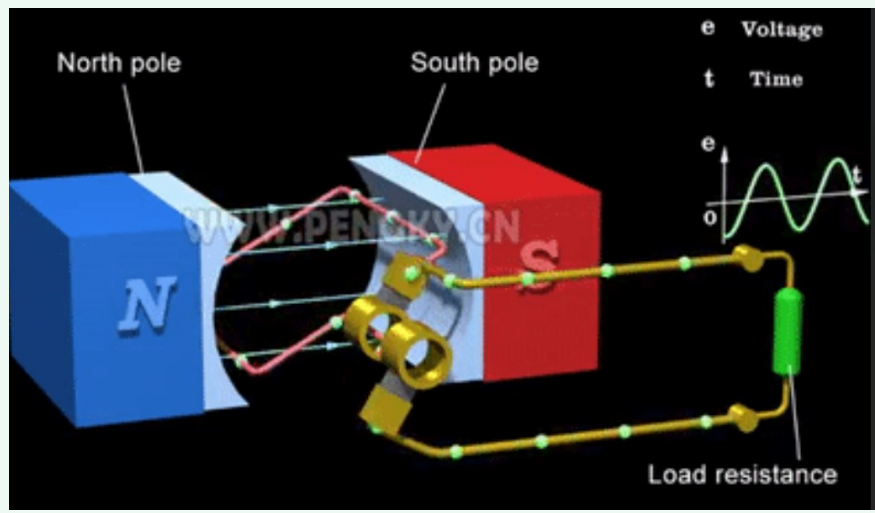
$$E(t) = E_{max} \sin(\omega t) \text{ with } E_{max} = N A B \omega$$

Consider the voltage $e(t)$ changing with time (t) . Induced by a AC generator.
 The voltage follows a sine function. The equation for a sine function is: $E(t) = E_{\text{max}} \sin(\omega t)$ with

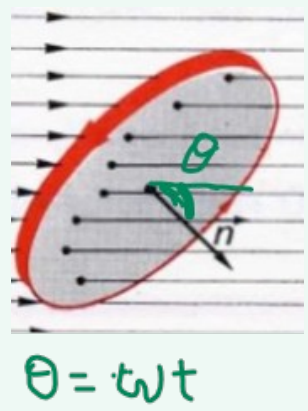
$\omega = 2\pi f = 2\pi/T$. ω is the angular frequency (rad/s)
 f is the frequency of the sine function (cycles per second).
 T is the period of the sine function (seconds per cycle). $T = 1/f$ or $f = 1/T$.

ω is also how fast the loop is spinning in the magnetic field in rad/s. with $2\pi \text{ rad} = 1 \text{ rotation} = 1 \text{ revolution} = 1 \text{ cycle}$. ω is the rotational speed or the rate of spin of the loop.

$E_{\text{max}} = N A B$ (N number of loops, A area of loop, B magnetic field, ω the rate of spin)



θ is angle between $(\vec{n}, \vec{B})$
 $\vec{n}$ is a unit vector perpendicular to A
 $\theta = \omega t \text{ (rad)}$



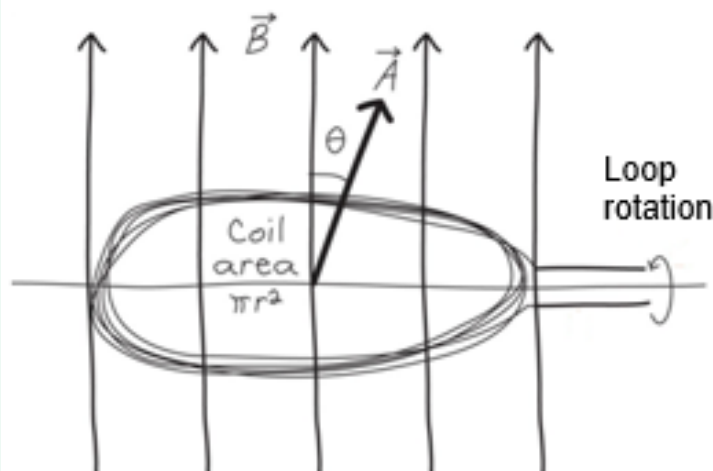
In the US $E_{\text{max}}=170\text{V}$
 $f = 60\text{Hz}$, $\omega = 2\pi (60)$
 $= 120 \pi \text{ rad/s}$

 $E(t) = 170 \sin(120\pi t)$

$E(t)$
 $= N A B \omega \sin(\omega t)$

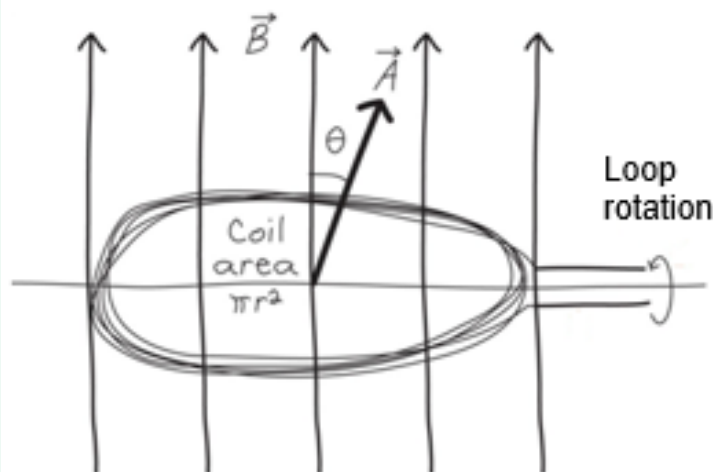
Example 27.5. Designing a Generator

An electric generator consists of a 100-turn circular coil 50 cm in diameter. It's rotated at $f = 60$ rev/s to produce standard 60 Hz alternating current. Find B needed for a peak output voltage of 170 V, which is the actual peak in standard 120 V household wiring.



Example 27.5. Designing a Generator

An electric generator consists of a 100-turn circular coil 50 cm in diameter. It's rotated at $f = 60$ rev/s to produce standard 60 Hz alternating current. Find B needed for a peak output voltage of 170 V, which is the actual peak in standard 120 V household wiring.



$$\Phi_{1 \text{ turn}} = B A \cos \theta = B \pi \left(\frac{d}{2} \right)^2 \cos(2\pi f t)$$

$$E = -\frac{d\Phi_B}{dt} = -N B \pi \left(\frac{d}{2} \right)^2 \frac{d}{dt} \cos(2\pi f t) \\ = 2N B \pi^2 f \left(\frac{d}{2} \right)^2 \sin(2\pi f t)$$

$$E_{\text{peak}} = 2N B \pi^2 f \left(\frac{d}{2} \right)^2$$

$$170 \text{ V} = 2(100) B \pi^2 (60 / \text{s}) \left(\frac{0.5 \text{ m}}{2} \right)^2$$

$$B = 23 \times 10^{-3} \text{ T}$$

Example 10 An Ac Generator

In Figure 22.21 the coil of the ac generator rotates at a frequency of $f = 60.0$ Hz and develops an emf of 120 V (rms; see Section 20.5). The coil has an area of $A = 3.0 \times 10^{-3} \text{ m}^2$ and consists of $N = 500$ turns. Find the magnitude of the magnetic field in which the coil rotates.

Reasoning The magnetic field can be found from the relation $\mathcal{E}_0 = NAB\omega$. However, in using this equation we must remember that $\mathcal{E}_0$ is the peak emf, whereas the given value of 120 V is not a peak value but an rms value. The peak emf is related to the rms emf by $\mathcal{E}_0 = \sqrt{2}\mathcal{E}_{\text{rms}}$, according to Equation 20.13.

Example 10 An Ac Generator

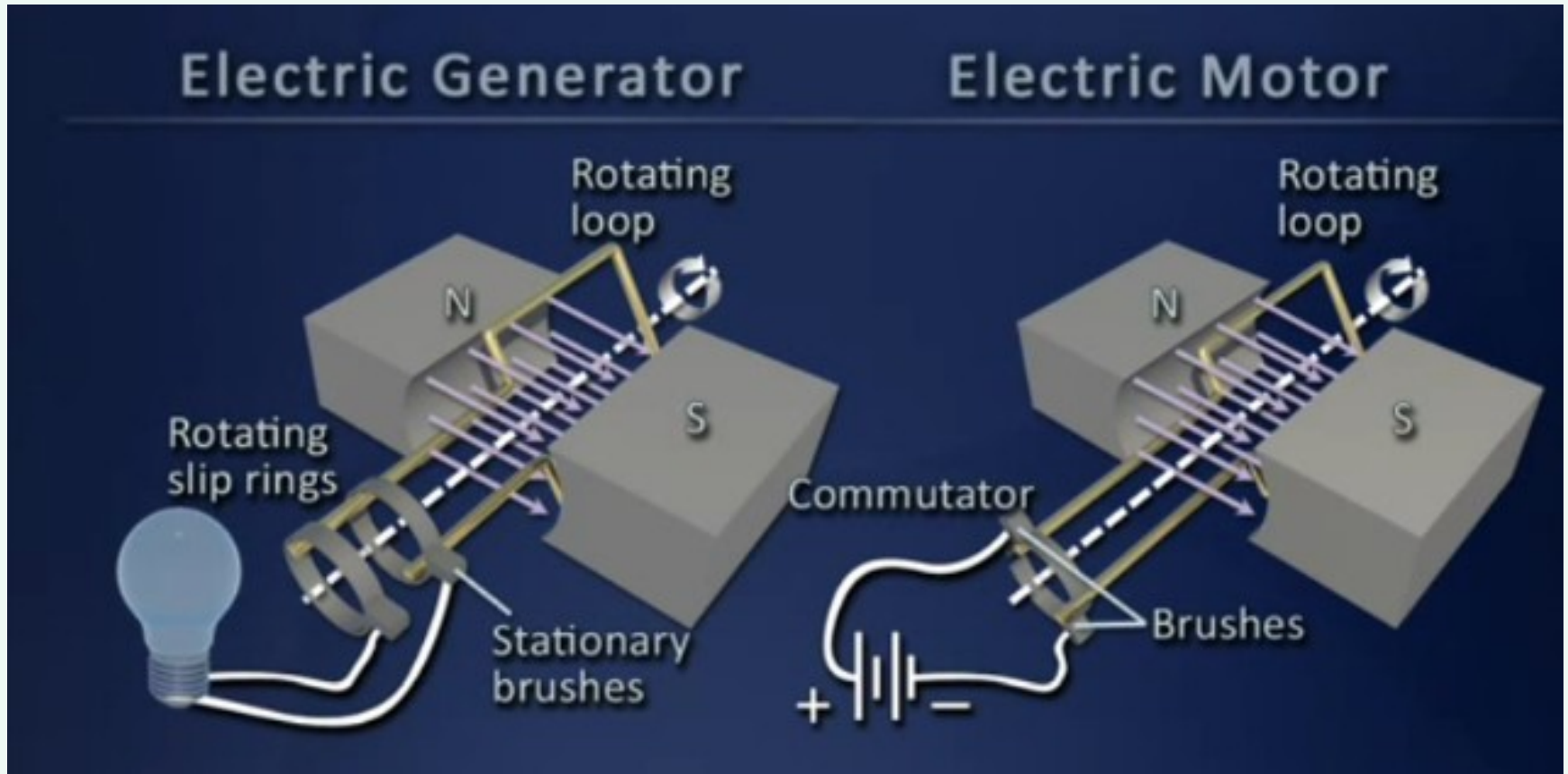
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Solution Solving $\mathcal{E}_0 = NAB\omega$ for B and using the facts that $\mathcal{E}_0 = \sqrt{2}\mathcal{E}_{\text{rms}}$ (Equation 20.13) and $\omega = 2\pi f$ (Equation 10.6), we find that the magnitude of the magnetic field is

$$B = \frac{\mathcal{E}_0}{NA\omega} = \frac{\sqrt{2}\mathcal{E}_{\text{rms}}}{NA2\pi f} = \frac{\sqrt{2} (120 \text{ V})}{(500)(3.0 \times 10^{-3} \text{ m}^2)2\pi(60.0 \text{ Hz})} = \boxed{0.30 \text{ T}}$$

Alternator/generator $\rightarrow$ mechanical energy $\rightarrow$ electrical energy.
motor $\rightarrow$ electrical energy $\rightarrow$ mechanical energy



Simulation AC generator (gifs) and generator jar. Discuss the dependence On the speed. Convert rot/min to rad/s. Guess the emf max. review rms. Use wolfram alpha.



Large scale power plant
(billion Watts)
Burn fuel -> steam -> turbine
-> electricity

turbines (100,000W)



Sponsored

30W Hand Crank Generator Manual Dynamo Power w/ DC-DC Converter For Laptop Phone

Brand New

\$119.99

or Best Offer

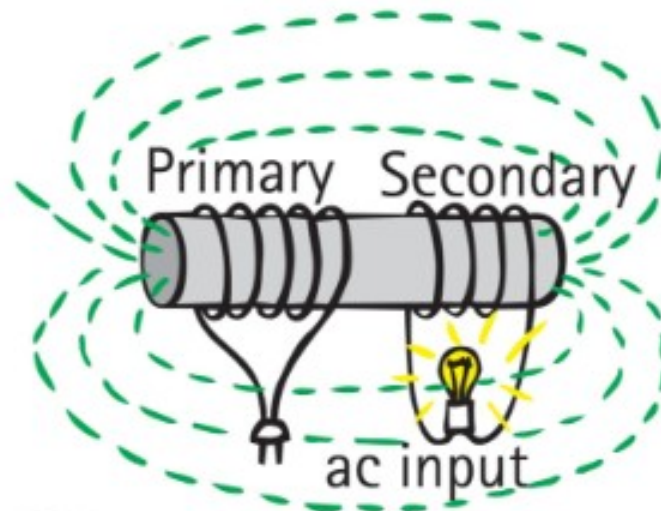
Free shipping

Free returns

13 watchers

Power Production, Continued-1

- Transformer



- Input coil of wire—the primary powered by ac voltage source
- Output coil of wire—the secondary connected to an external circuit

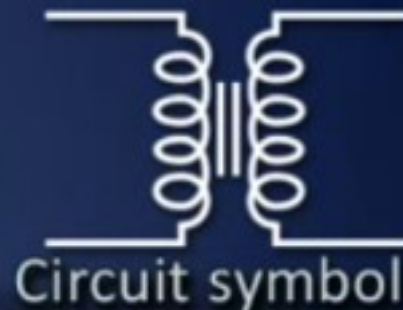
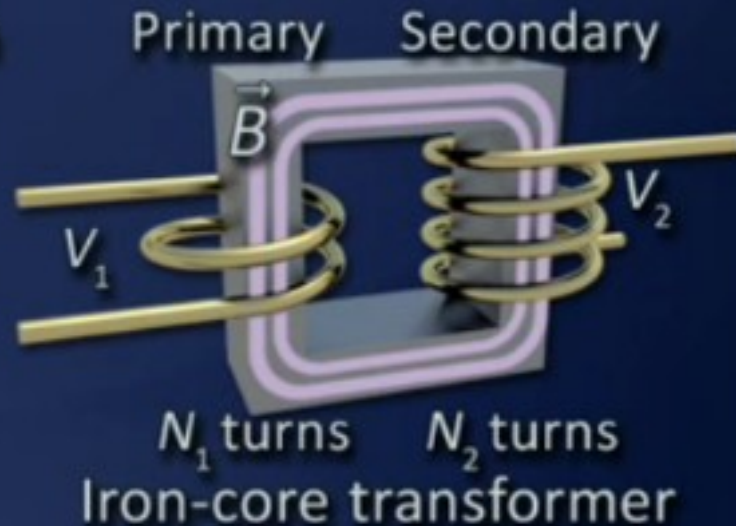
<https://www.youtube.com/watch?v=GMePE7NZcxw>

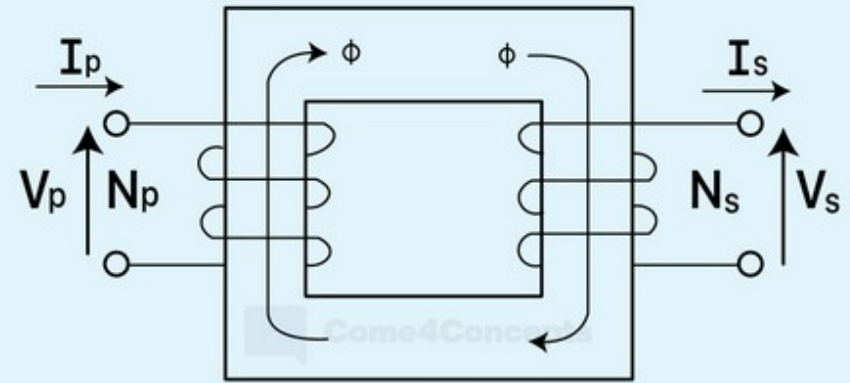
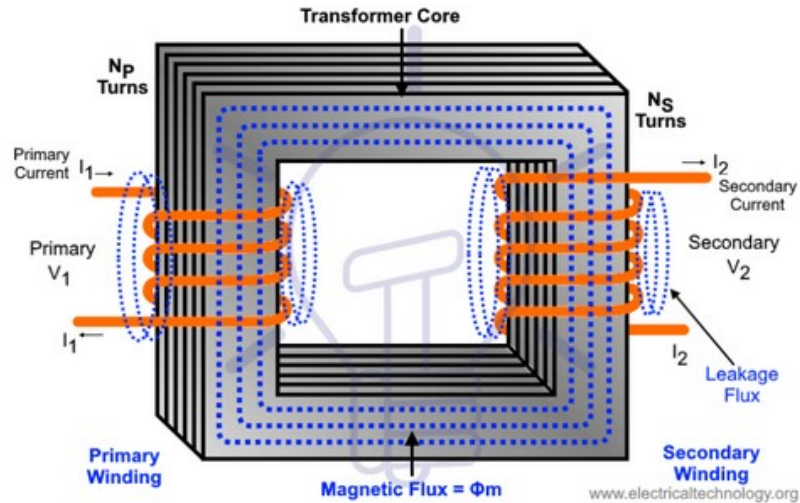
Step up transformers or step down transformers. Demo.

Transformers

- Change voltage levels with alternating current
- Primary coil produces changing magnetic flux
- Induces voltage in secondary coil
- Iron core concentrates magnetic flux

$$V_2 = \left(\frac{N_2}{N_1} \right) V_1$$

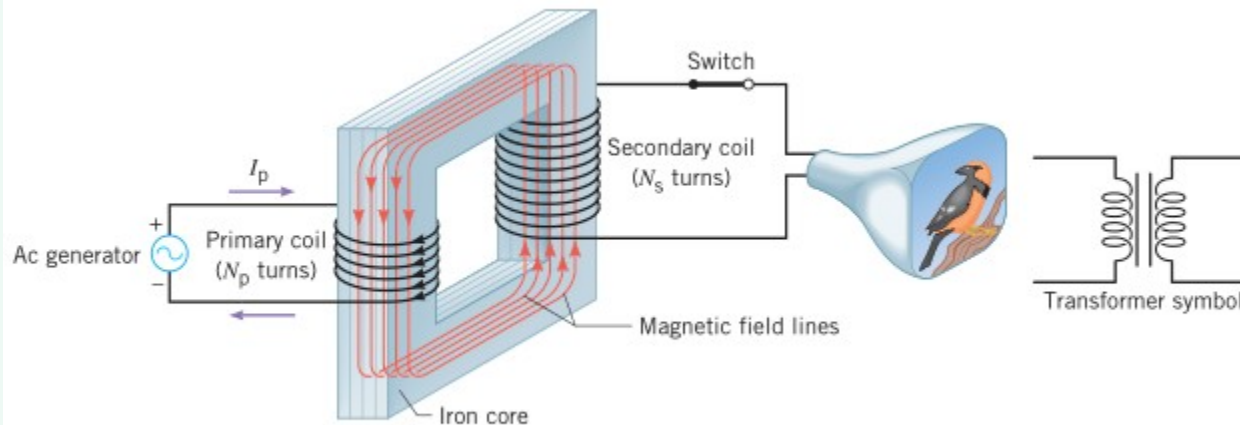


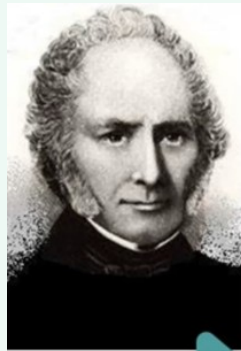


$\phi = \text{Constant}$

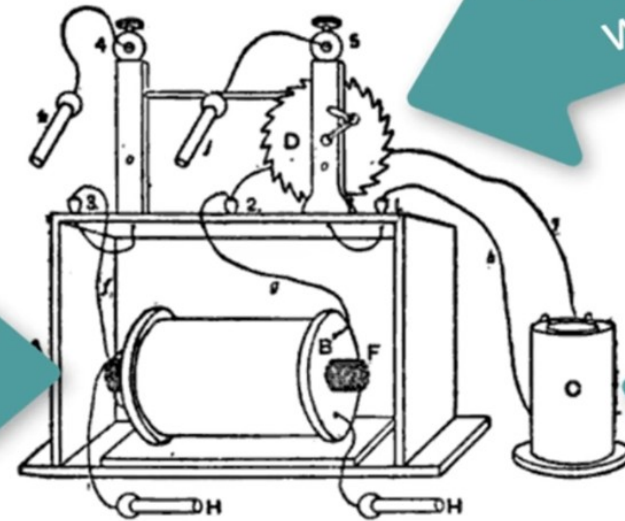
$$\frac{V_s}{V_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p}$$

Ideal Transformer



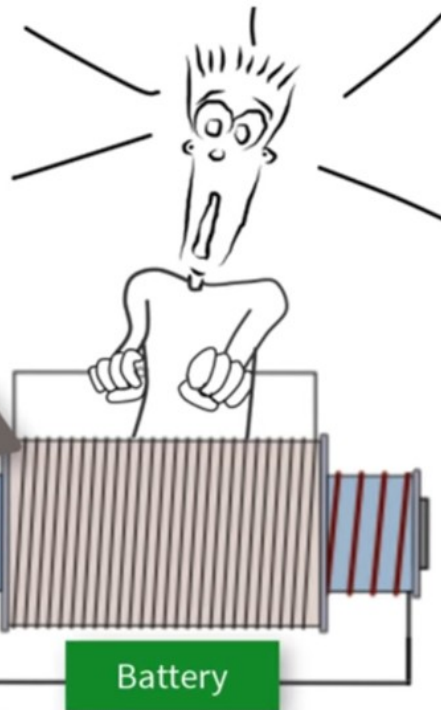


coils



wheel

battery



Secondary Coil

Primary Coil

Battery

William Sturgeon

First electromagnet.
Improved motor.



Be aware of “pseudo” medical sciences.

ELECTRICITY A NATURAL CURE

NO MORE USE FOR CRUTCHES OR DRUGS.

TAKE A TREATMENT FREE!

I have so adapted my application of medical electricity that I positively cure Rheumatism, Kidney and Stomach Troubles and all affections of the vital organs. To illustrate the remedial power of my

Dr. McLaughlin's Belt

I will give you a free treatment in my office by the galvanic current from my grand stationary battery. This is a breezy and exhilarating test of that curative power now used the world over.

To cure your ailment I will fit you with one of my famous Belts, which you wear at night until your system becomes charged with the reviving voltage. This keeps up the action of the blood vessels, contracts and strengthens relaxed muscles and glands. It casts out impurities, which are the cause of pain, and restores to the body its wonted energy. You feel the glow, the enlivening sensation of this power from the start. The improvements my Belt possesses above all others are recognized and appreciated by the thousands who are using my appliances. The cures I have made, the testimony from all quarters, the comfort of wearing my Belt, the method of adjusting the power, its thoroughly curative work, constitute it the pre-eminent remedy for the severest pain or the most trying weakness.

Call and take a treatment free from my Static Battery and test my Belt, or let me send you my new book, free.

← Rheumatism and Kidney Trouble →



Example 14 A Step-Down Transformer

A step-down transformer inside a stereo receiver has 330 turns in the primary coil and 25 turns in the secondary coil. The plug connects the primary coil to a 120-V wall socket, and there is a current of 0.83 A in the primary coil while the receiver is turned on. Connected to the secondary coil are the transistor circuits of the receiver. Find (a) the voltage across the secondary coil, (b) the current in the secondary coil, and (c) the average electric power delivered to the transistor circuits.

Reasoning The transformer equation, Equation 22.12, states that the secondary voltage V_s is equal to the product of the primary voltage V_p and the turns ratio N_s/N_p . On the other hand, Equation 22.13 indicates that the secondary current I_s is equal to the product of the primary current I_p and the inverse turns ratio N_p/N_s . The average power delivered to the transistor circuits is the product of the secondary current and the secondary voltage.

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Solution (a) The voltage across the secondary coil can be found from the transformer equation:

$$V_s = V_p \frac{N_s}{N_p} = (120 \text{ V}) \left(\frac{25}{330} \right) = \boxed{9.1 \text{ V}} \quad (22.12)$$

(b) The current in the secondary coil is

$$I_s = I_p \frac{N_p}{N_s} = (0.83 \text{ A}) \left(\frac{330}{25} \right) = \boxed{11 \text{ A}} \quad (22.13)$$

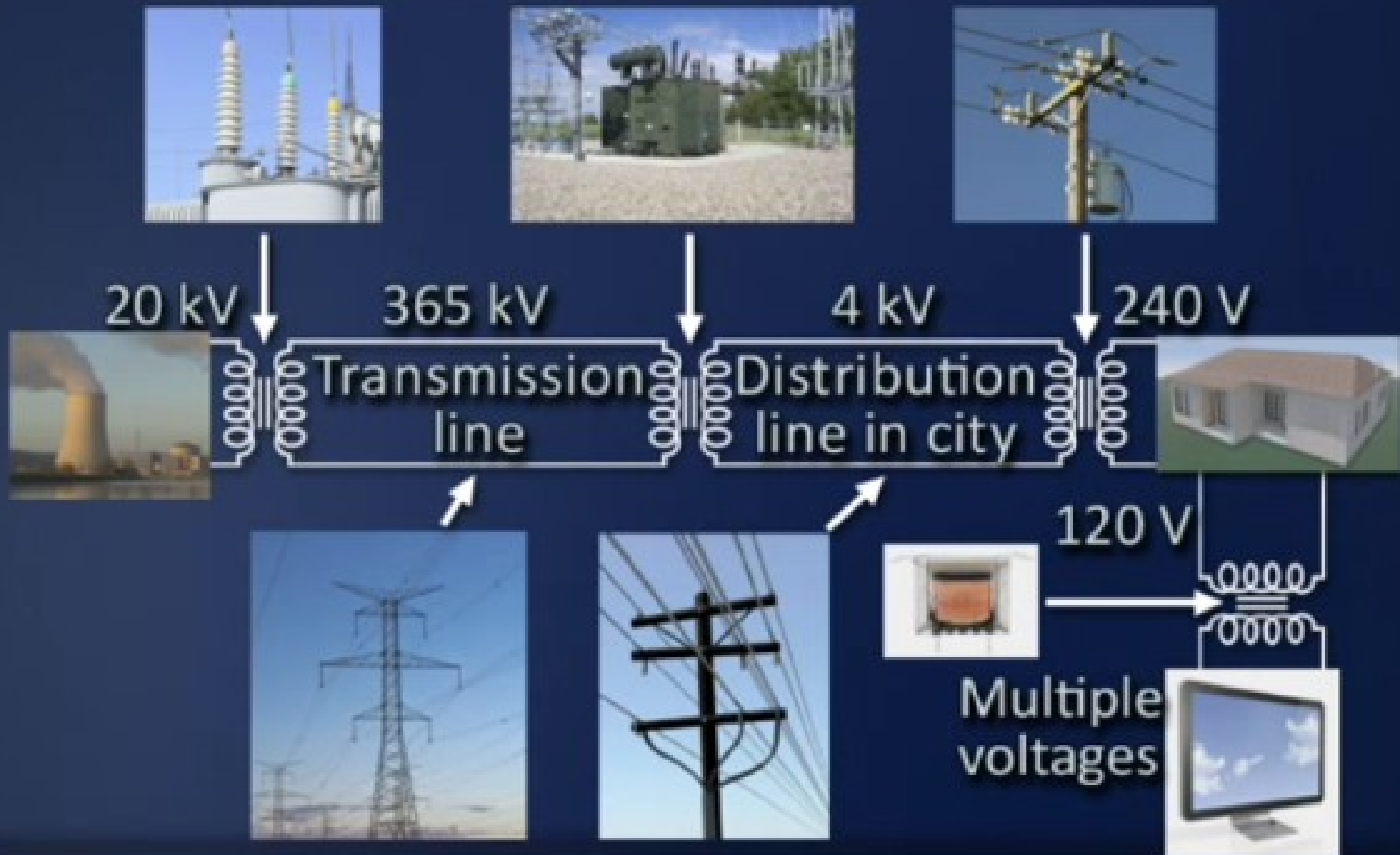
(c) The average power $\bar{P}_s$ delivered to the secondary coil is the product of I_s and V_s :

$$\bar{P}_s = I_s V_s = (11 \text{ A})(9.1 \text{ V}) = \boxed{1.0 \times 10^2 \text{ W}} \quad (20.15a)$$

As a check on our calculation, we verify that the power delivered to the secondary coil is the same as that sent to the primary coil from the wall receptacle:

$$\bar{P}_p = I_p V_p = (0.83 \text{ A})(120 \text{ V}) = 1.0 \times 10^2 \text{ W}$$

Transformers in the Grid



ORIGINAL experiment of Michael Faraday → transformer

Here is a diagram showing the basic set-up of his experiment.

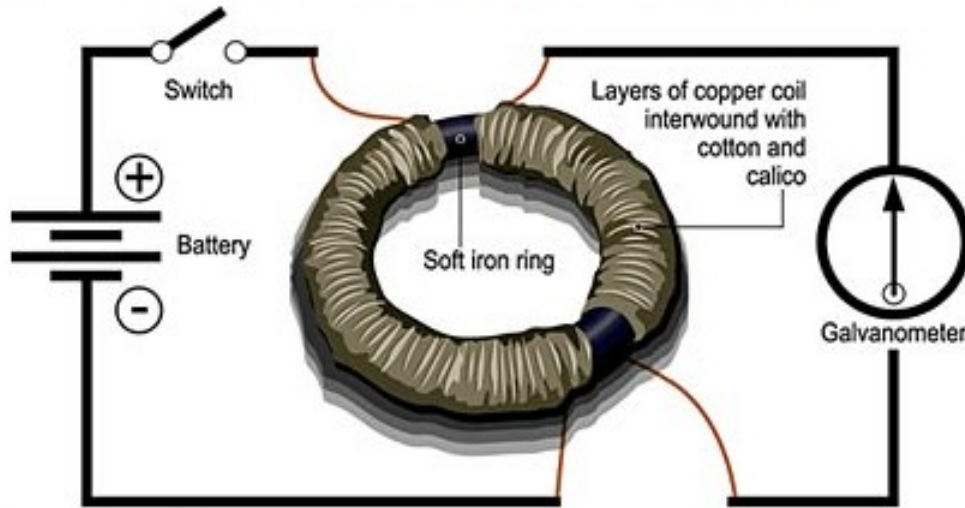


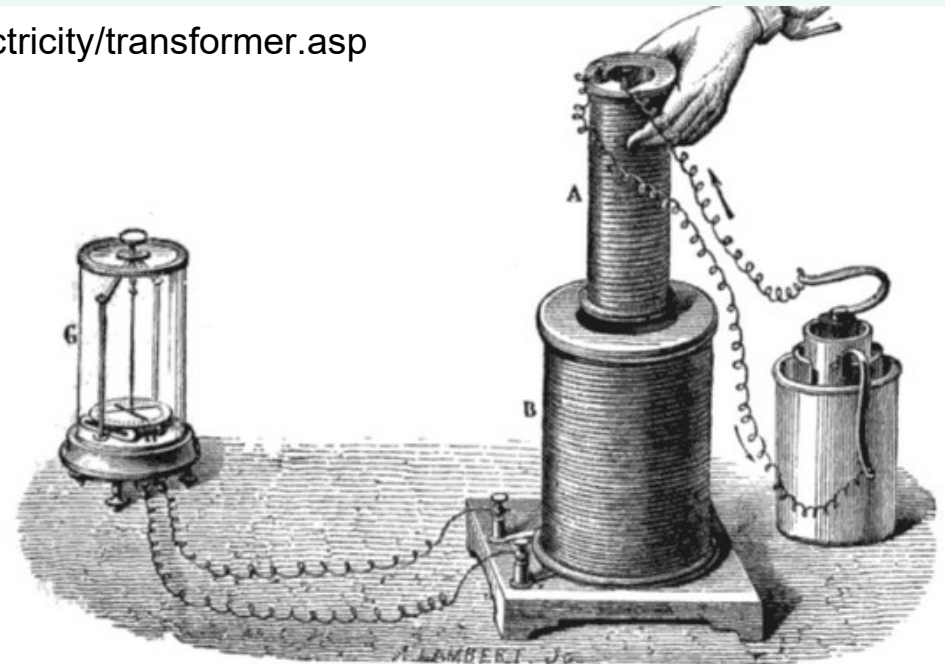
Illustration : Paul Weston

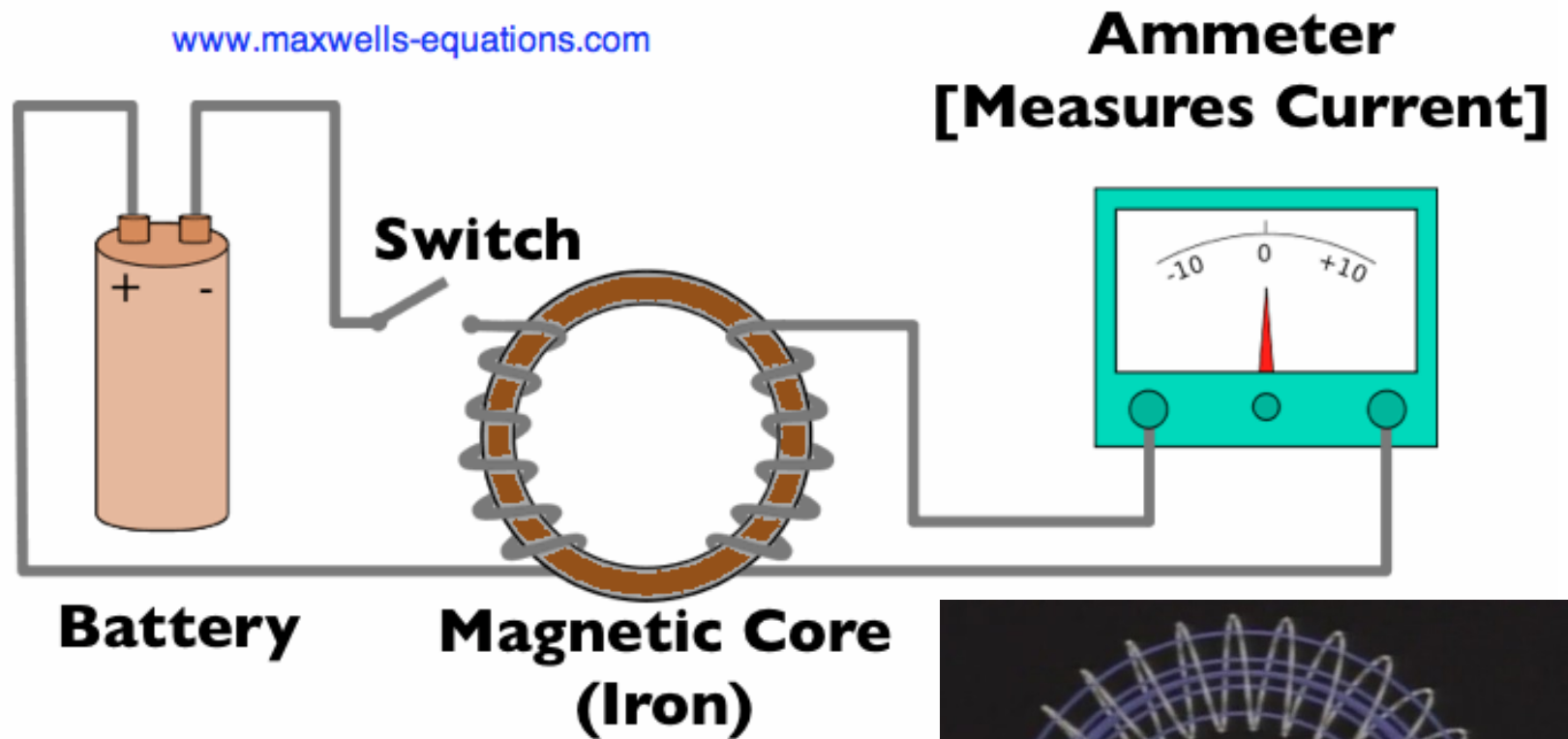
Remember the demo
With the screw driver.

The magnetic field lines are
guided/trapped in the core.

See next slide To understand.

<http://www.engineering-timelines.com/how/electricity/transformer.asp>

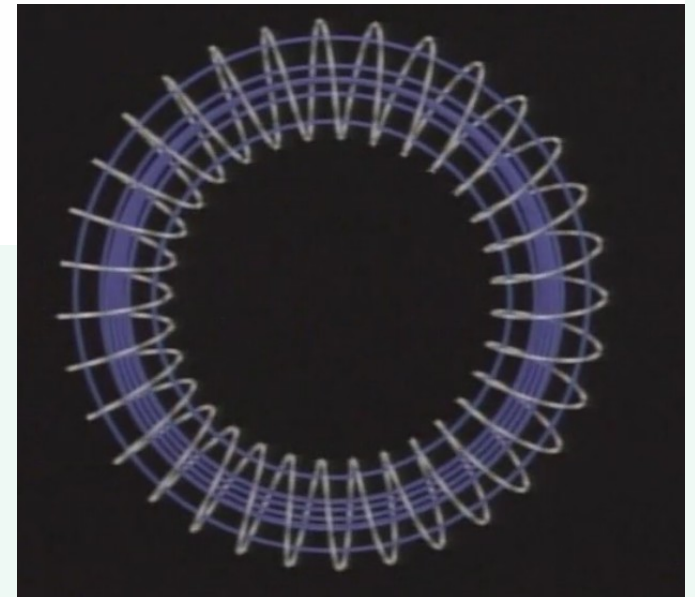




See link to website with java sim

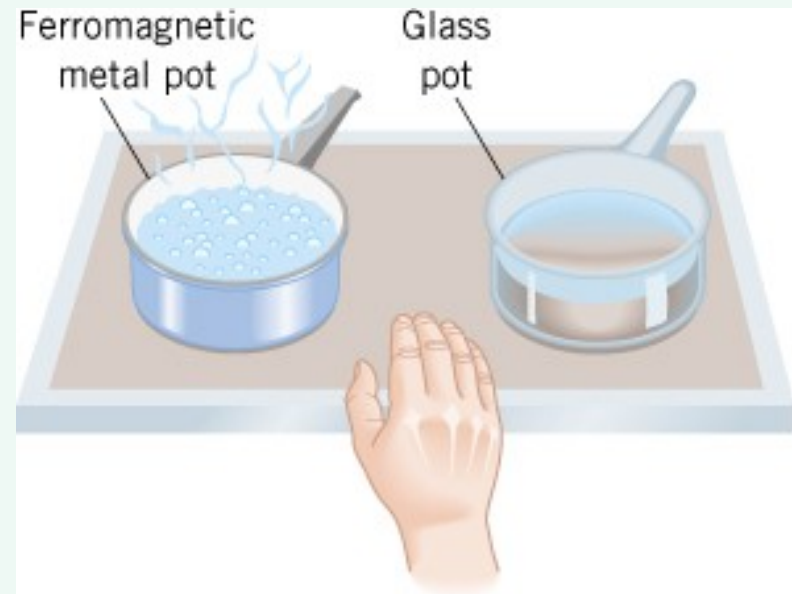
B inside a toroid is

$$B = \frac{\mu_0 N I}{2\pi r} \text{ (within the toroid).}$$



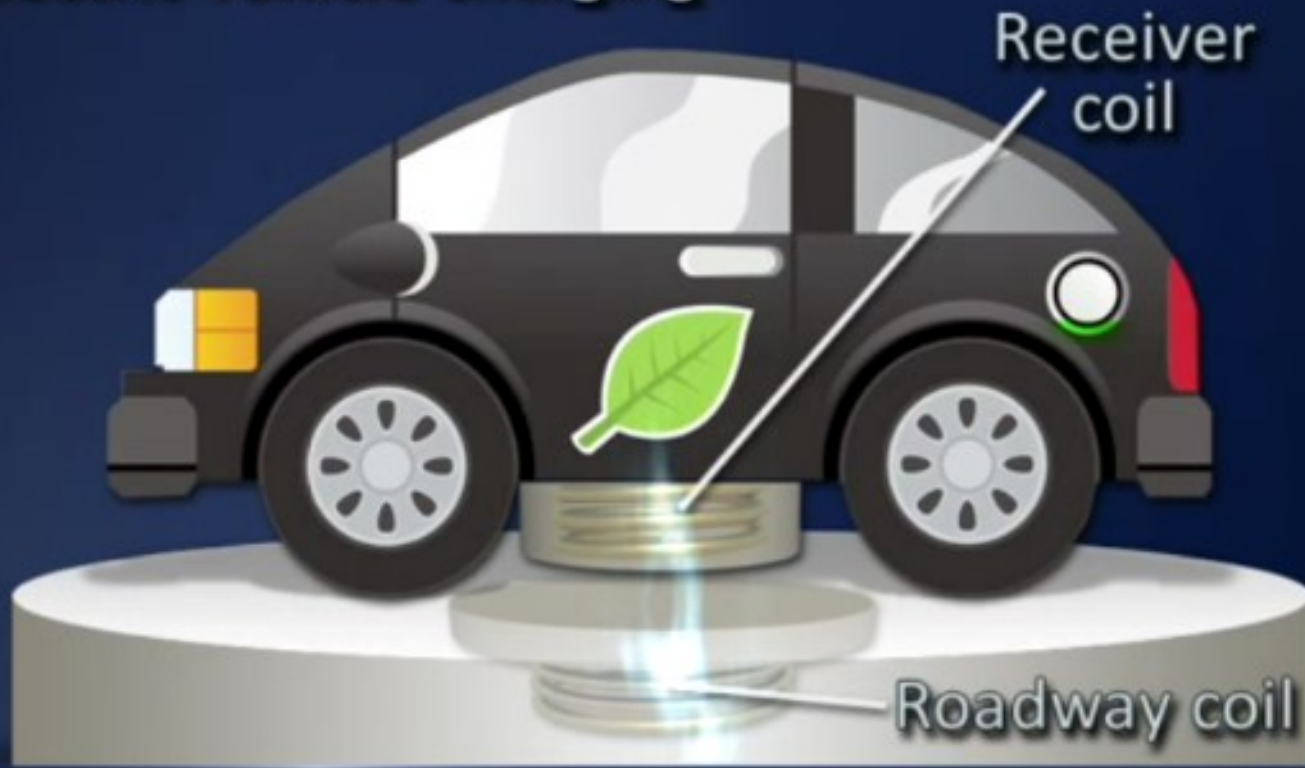
***A changing flux induced current in bulk conductor
(does not have to be a wire). Example: An Induction Stove***

Two pots of water are placed on an induction stove at the same time. The stove itself is cool to the touch. The water in the ferromagnetic metal pot is boiling while that in the glass pot is not. How can such a cool stove boil water, and why isn't the water in the glass pot boiling?



Inductive Power Transfer

Electric vehicle charging



Alternating
current

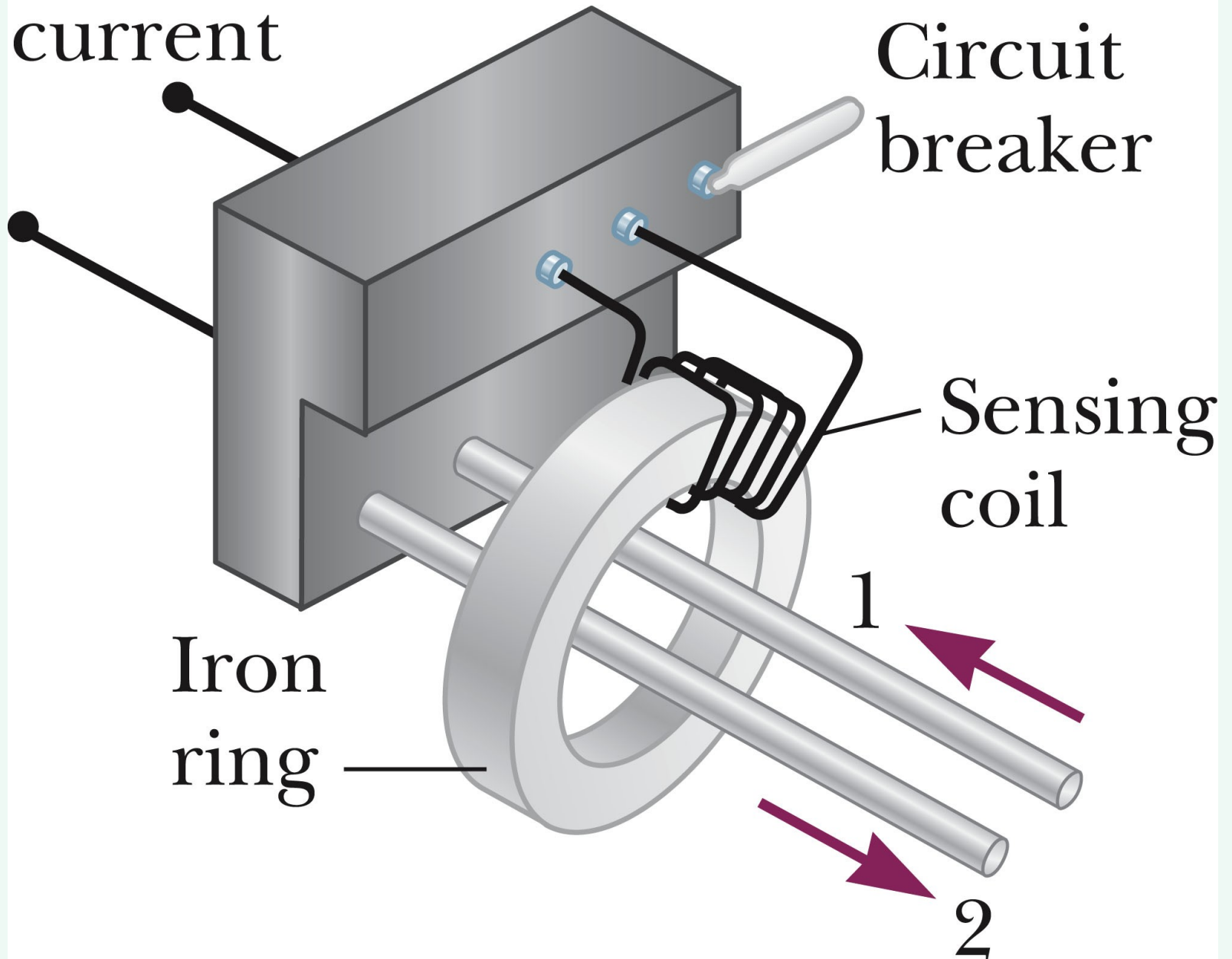
Circuit
breaker

Sensing
coil

Iron
ring

1

2



Magnetized
portion of
string

N S

Pickup
coil

N

S

Magnet

To amplifier

Guitar
string

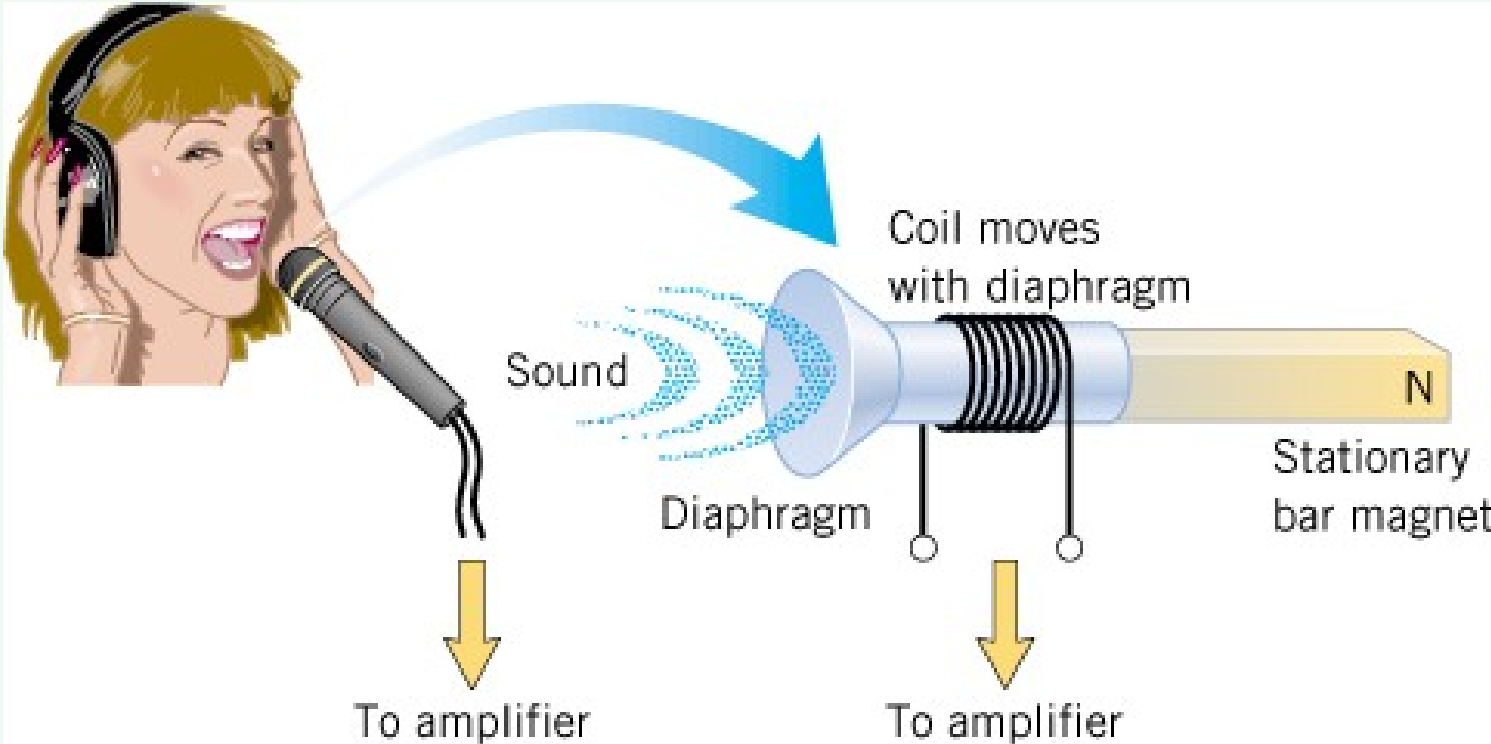


b

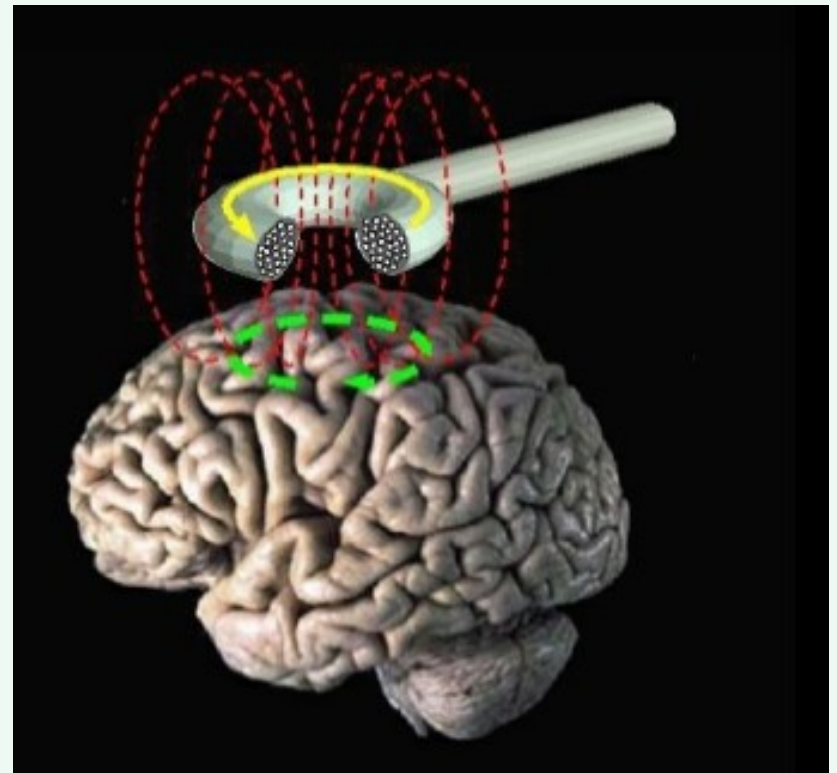
a

22.6 Applications of Electromagnetic Induction to the Reproduction of Sound

inductive inducers. This is dynamic microphone (now capacitive microphone)



Transcranial magnetic stimulation



Transcranial magnetic stimulation (TMS) is a noninvasive form of brain stimulation in which a changing magnetic field is used to cause electric current at a specific area of the brain through electromagnetic induction. An electric pulse generator, or stimulator, is connected to a magnetic coil, which in turn is connected to the scalp. The stimulator generates a changing electric current within the coil which induces a magnetic field; this field then causes a second inductance of inverted electric charge within the brain itself.[1]: 3[2]

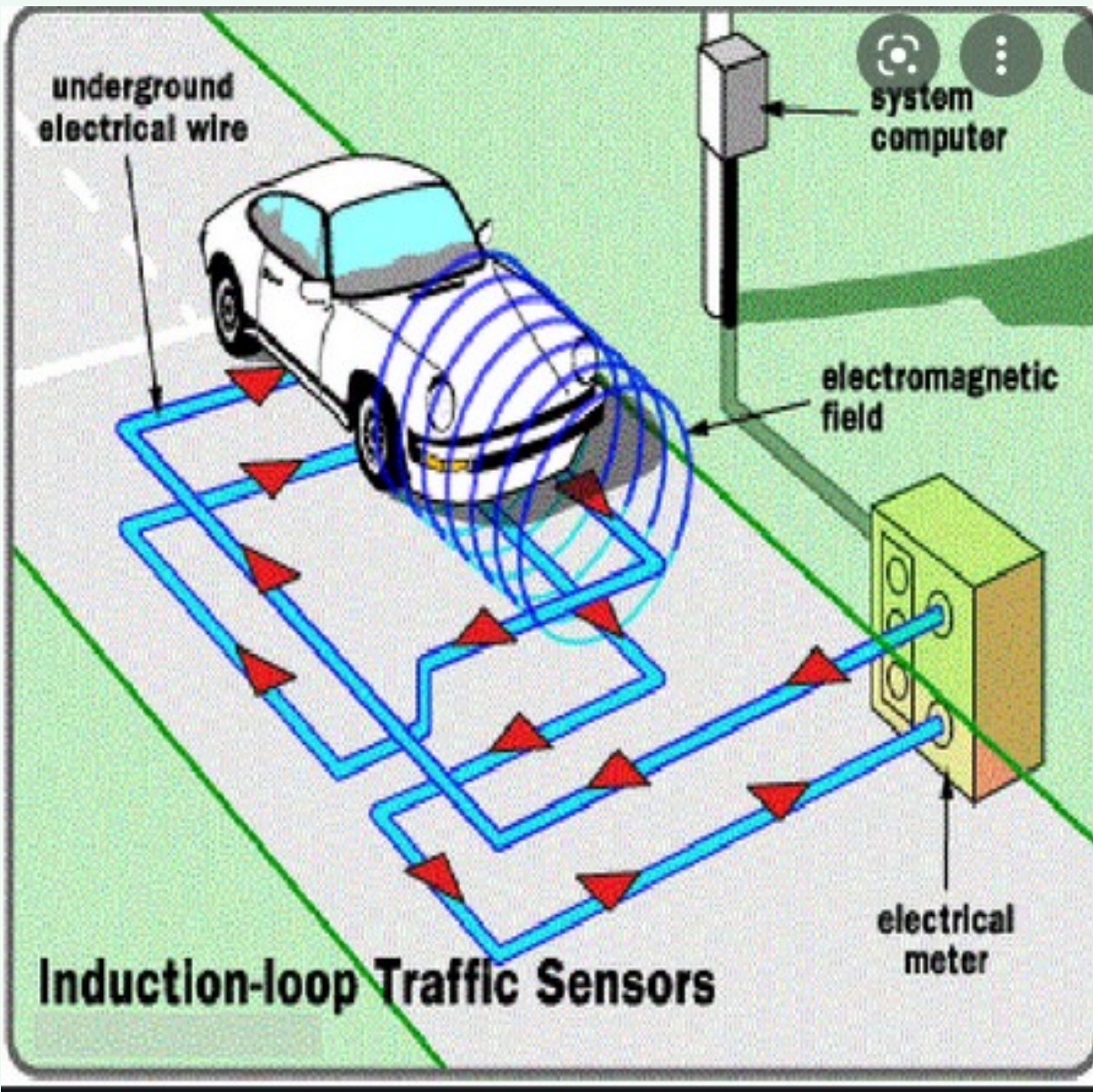
TMS has shown diagnostic and therapeutic potential in the central nervous system with a wide variety of disease states in neurology and mental health, with research still evolving

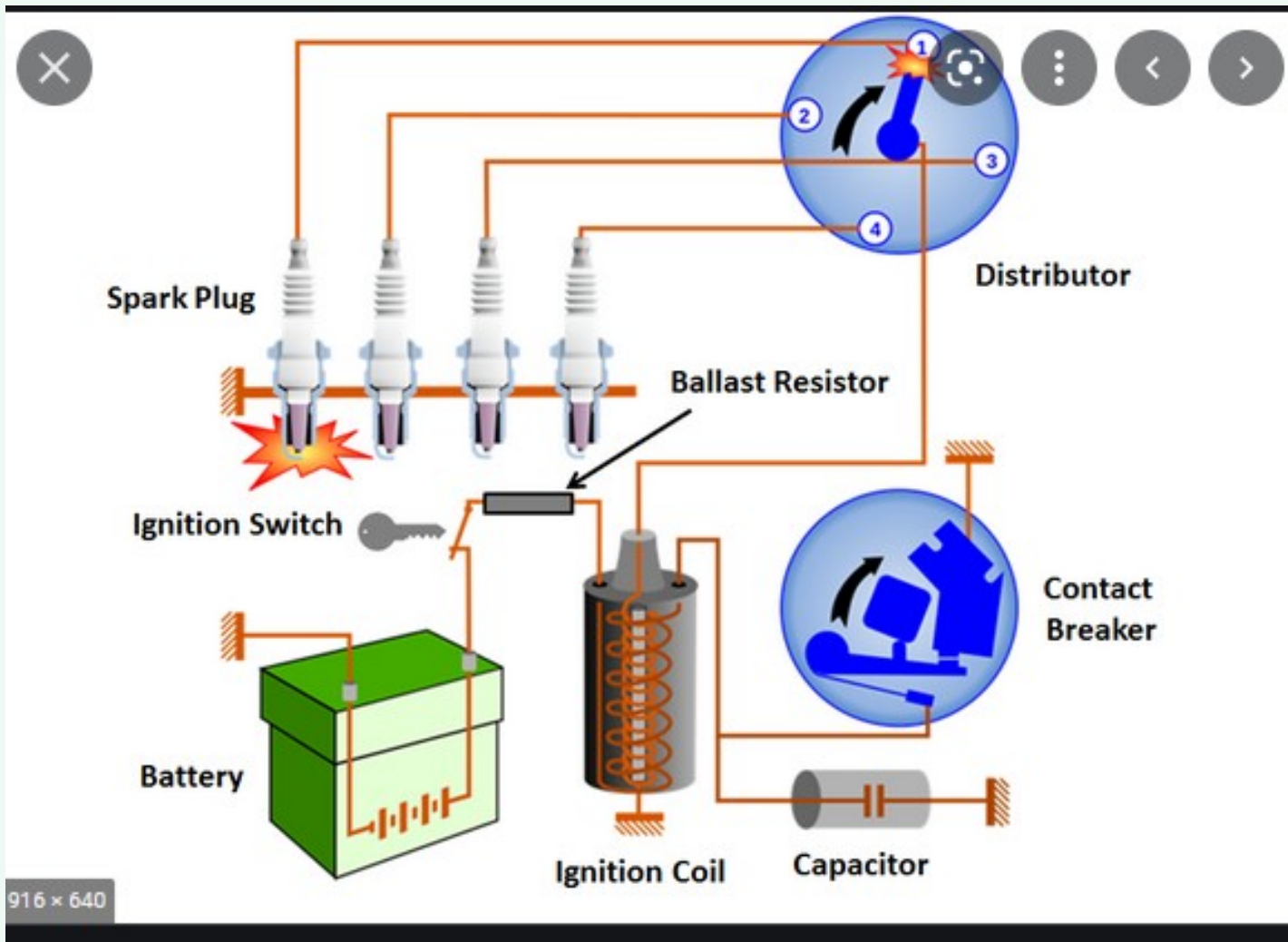
Airport Security

Pulsed-induction metal detector

- Current pulse generates brief magnetic field
- Pulse ends
- Field collapses
- Changing field induces “reflected” pulse
- Eddy currents in metal slow the field collapse

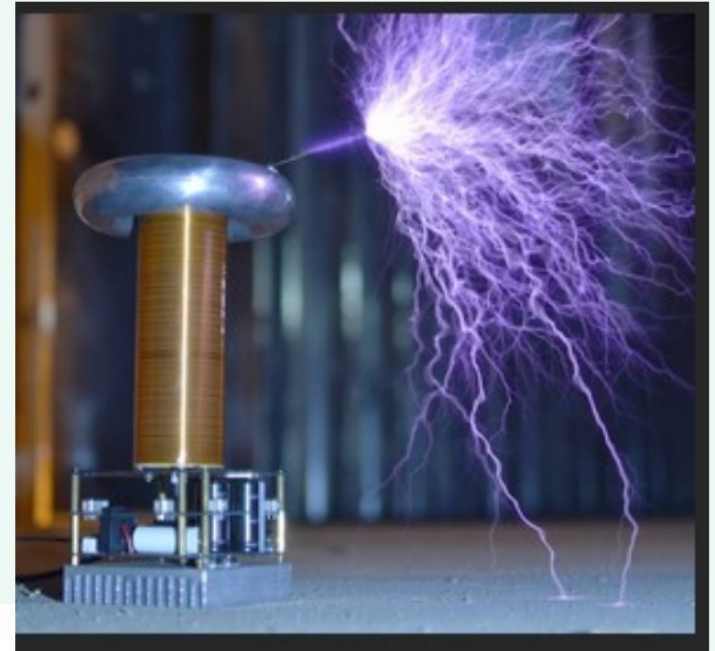
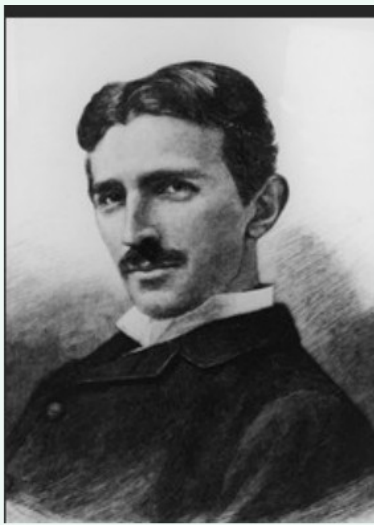




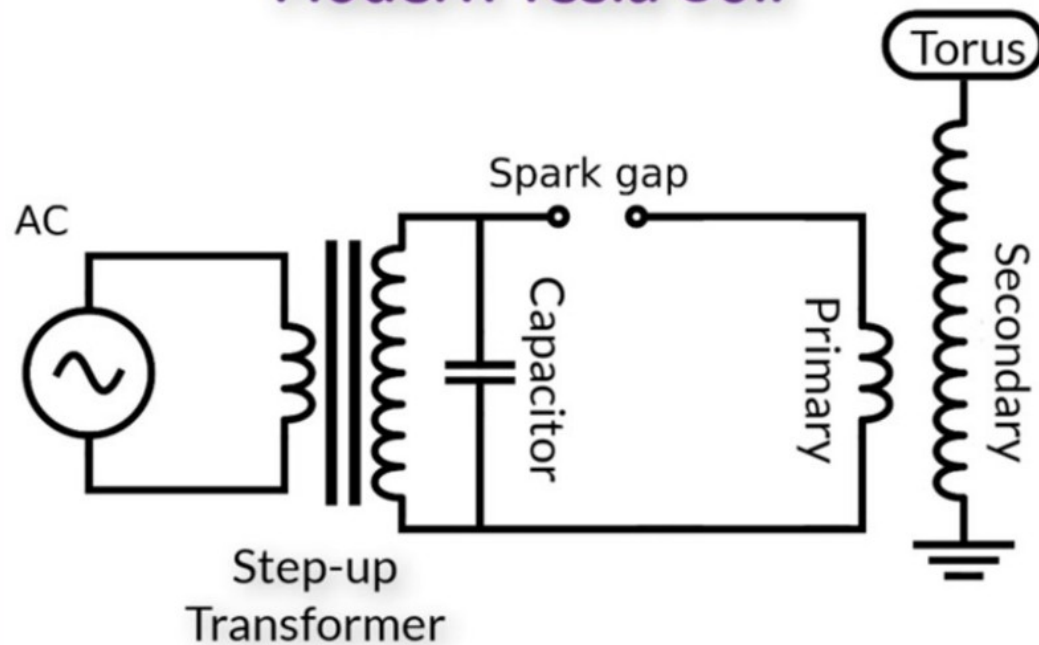


https://www.youtube.com/watch?v=0yK3Opq_i0M

<https://www.youtube.com/watch?v=OMLSNwQiiKg>



Modern Tesla Coil

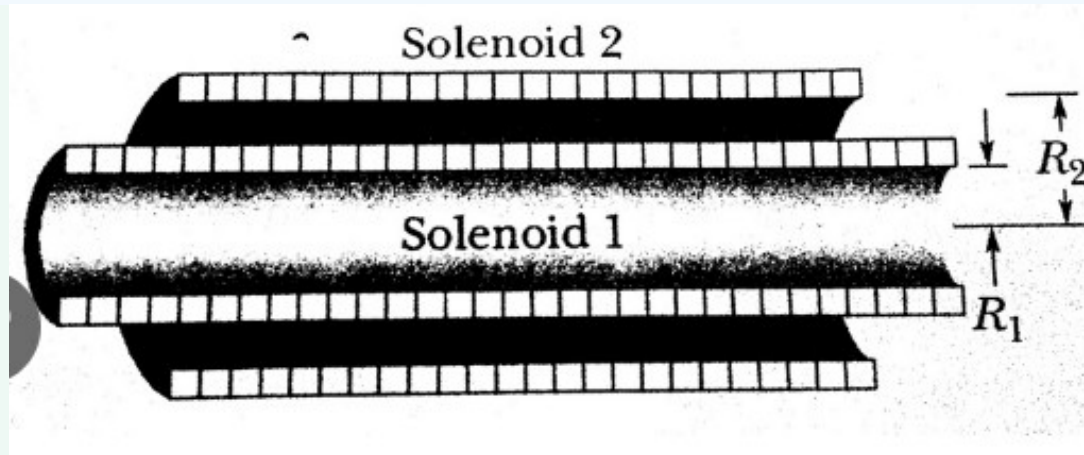


**Marconi took this idea
To make radio wave
And got the Nobel Prize.
Tesla died penny less.**

Tesla → AC , wireless tech

More about solenoids

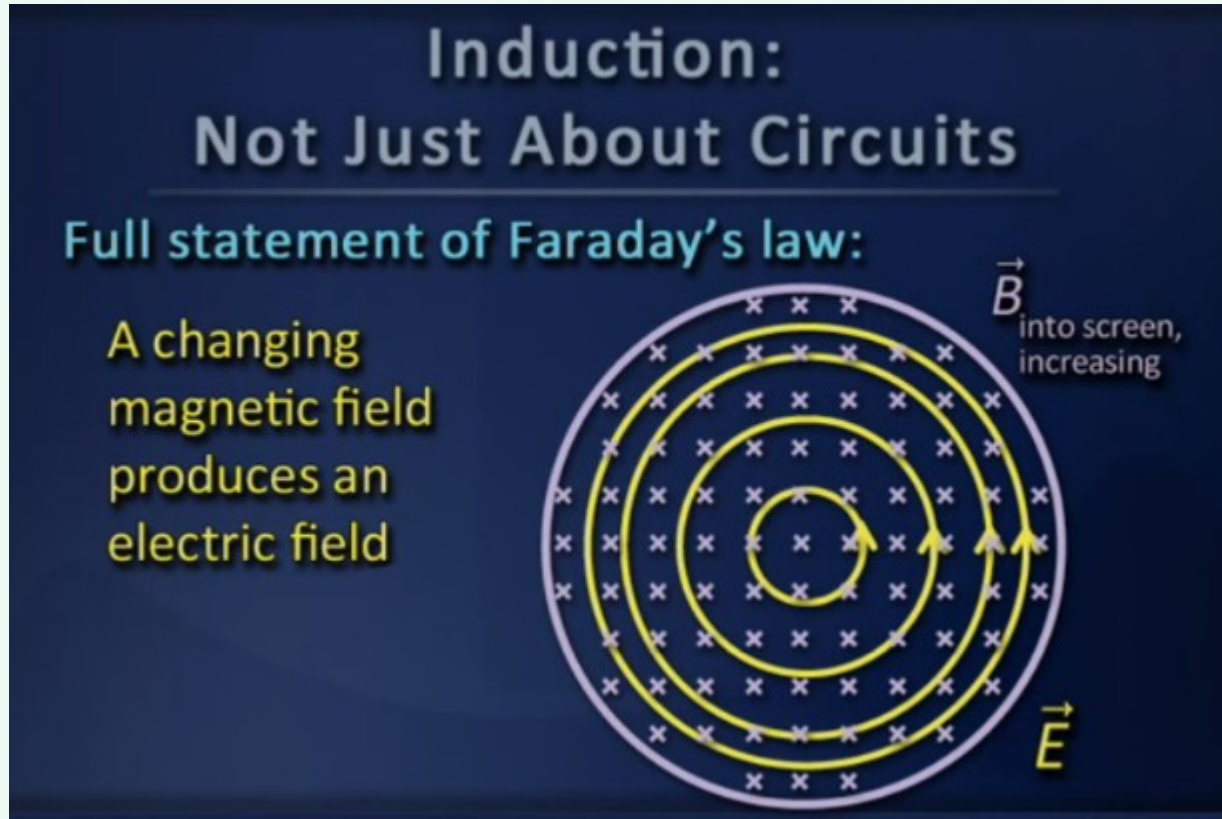
The current through the windings of a solenoid with $n = 2000$ turns per meter is changing at a rate $dI/dt = 3.0 \text{ A/s}$. (See [Sources of Magnetic Fields](#) for a discussion of solenoids.) The solenoid is 50-cm long and has a cross-sectional diameter of 3.0 cm. A small coil consisting of $N = 20$ closely wound turns wrapped in a circle of diameter 1.0 cm is placed in the middle of the solenoid such that the plane of the coil is perpendicular to the central axis of the solenoid. Assuming that the infinite-solenoid approximation is valid at the location of the small coil, determine the magnitude of the emf induced in the coil.



Reference:

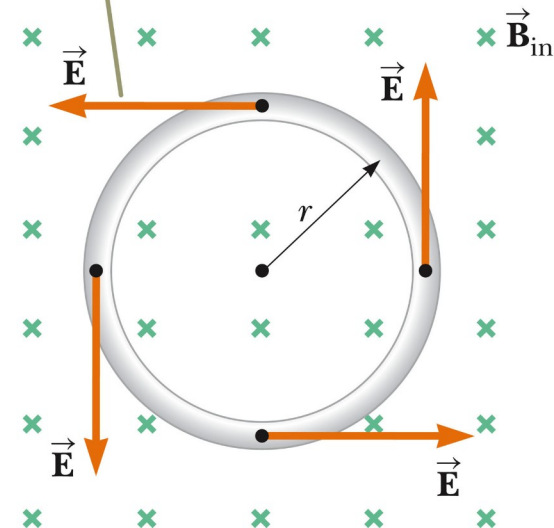
[https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_\(OpenStax\)/Book%3A_University_Physics_II_-_Thermodynamics_Electricity_and_Magnetism_\(OpenStax\)](https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_(OpenStax)/Book%3A_University_Physics_II_-_Thermodynamics_Electricity_and_Magnetism_(OpenStax))

An electric field is induced in space even if there is no conductor.
 The electric field circulate. The magnitude depends on the rate of change of the flux. The electric field is stronger on the outside here.



$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

If $\vec{B}$ changes in time, an electric field is induced in a direction tangent to the circumference of the loop.

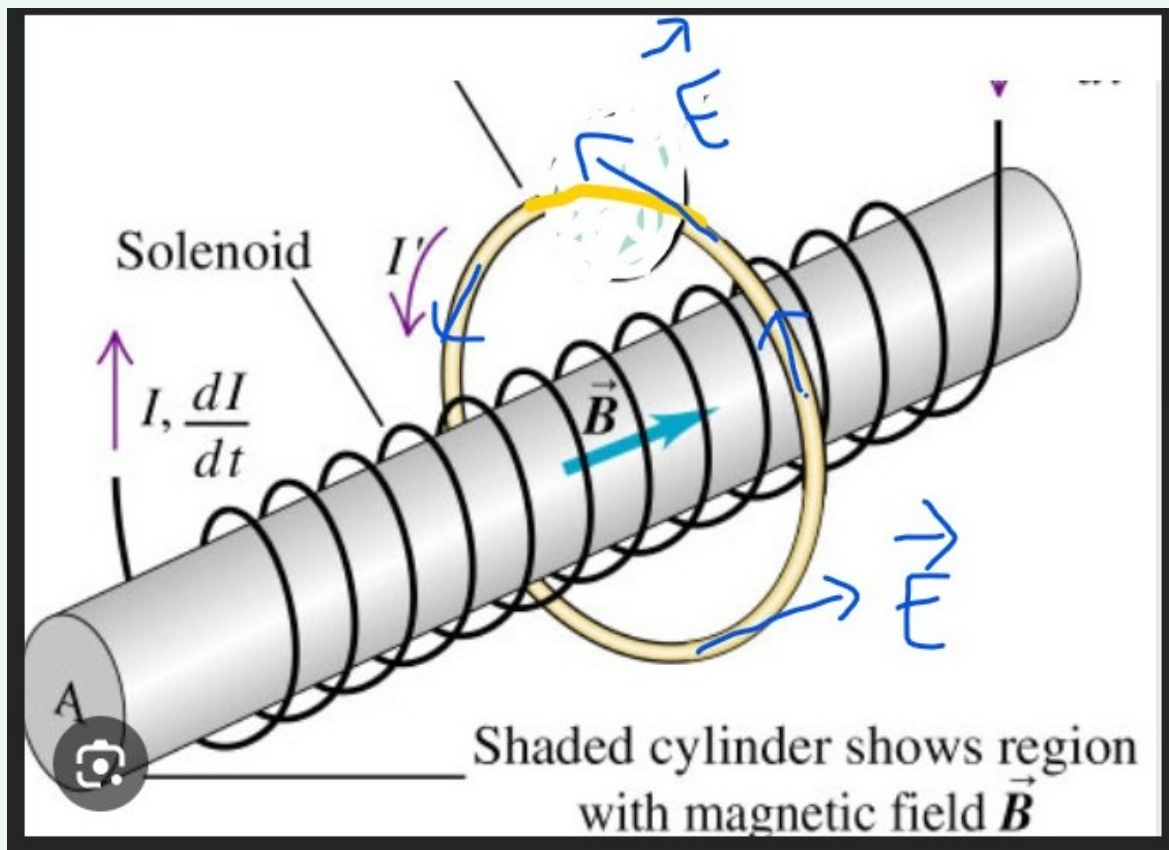


If there is a conductive loop. Kirchhoff law does not work any more! The electric field curls around itself.

Video maxwell in next unit

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint \vec{E} \cdot d\vec{l} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$



$$\sum \vec{E}_{//} \Delta l = \mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t}$$

A long solenoid (radius = 3.0 cm, 2500 turns per meter) carries a current given by $I = 0.30 \sin(200 t)$ A, where t is measured in s. When $t = 2.5$ ms, what is the magnitude of the induced electric field at a point which is 4.0 cm from the axis of the solenoid?

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

The magnetic field inside the solenoid is $B = \mu_0 n I$, so $\frac{dB}{dt} = \mu_0 n \frac{dI}{dt}$. Compute $\frac{dI}{dt}$: $I = 0.30 \sin(200t)$, $\frac{dI}{dt} = 0.30 \times 200 \cos(200t) = 60 \cos(200t)$. At $t = 0.0025$ s, $200t = 0.5$ rad, $\cos(0.5) \approx 0.8776$, so $\frac{dI}{dt} = 60 \times 0.8776 = 52.66$ A/s. Then, $\frac{dB}{dt} = (4\pi \times 10^{-7}) \times 2500 \times 52.66 = 0.1657$ T/s.

For $r > R$, the induced electric field outside is $E = \frac{\frac{dB}{dt} R^2}{2r}$. So, $E = \frac{0.1657 \times (0.03)^2}{2 \times 0.04} = \frac{0.1657 \times 0.0009}{0.08} = 1.86 \times 10^{-3}$ V/m.

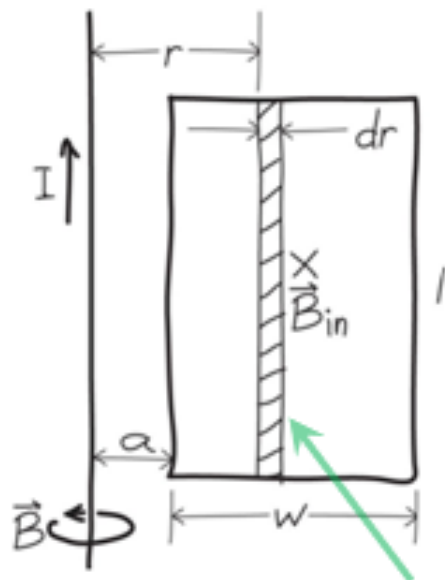
Calculus part

Example 27.2. Nonuniform Field

A long, straight wire carries current I .

A rectangular wire loop of dimensions l by w lies in a plane containing the wire, with its closest edge a distance a from the wire, and its dimension l parallel to the wire.

Find the magnetic flux through the loop.



Area element for integration

$$\begin{aligned}\Phi_B &= \int \mathbf{B} \cdot d\mathbf{A} = \int_a^{a+w} \frac{\mu_0 I}{2\pi r} l \, dr = \frac{\mu_0 I l}{2\pi} \int_a^{a+w} \frac{dr}{r} \\ &= \frac{\mu_0 I l}{2\pi} \ln \frac{a+w}{a}\end{aligned}$$

More about solenoids

Area A , length l , Number loops N

Back emf when B expands or collapses.
(show sims)

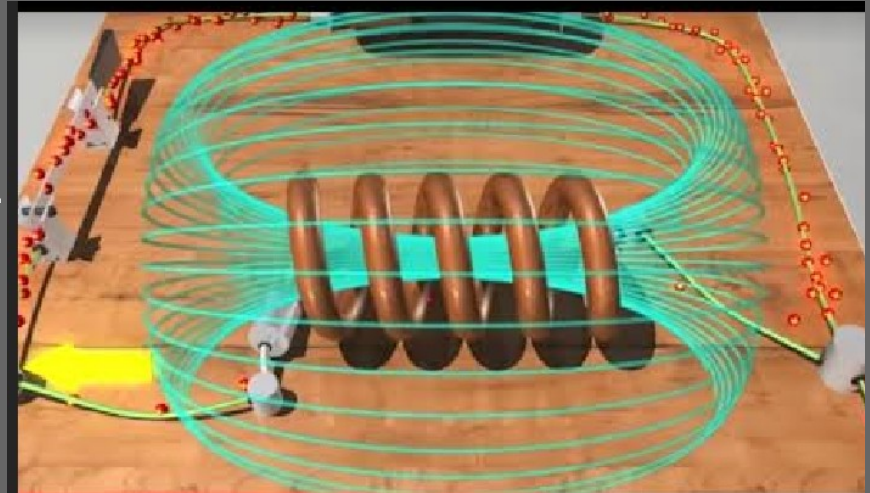
Back emf is given by Faradays law:
 $\text{Emf} = d\phi/dt$

And $B = \mu_0 n I$ for ideal solenoid. $n = N/l$
 $\Phi = N B A = \mu_0 N^2 A i / l$

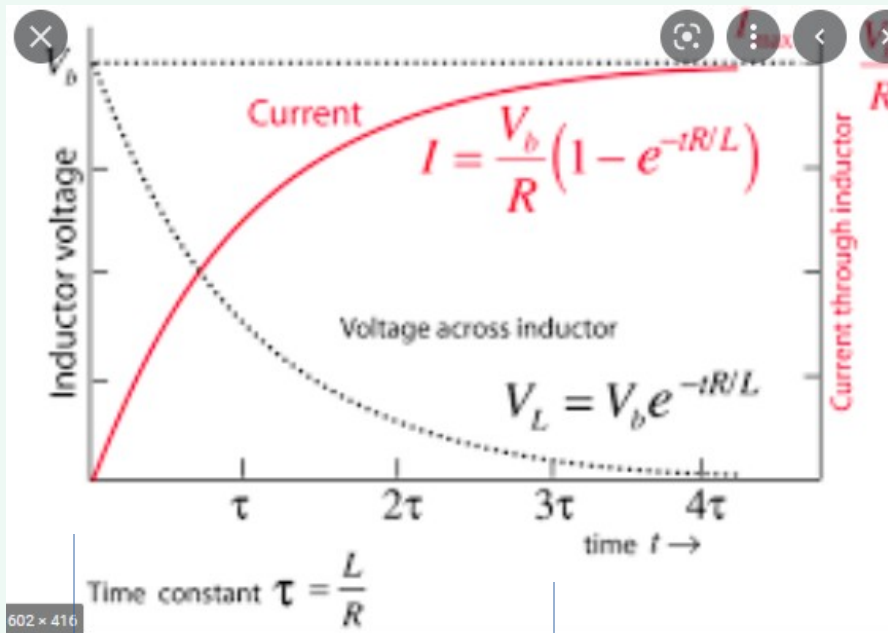
$\text{Emf} = d\phi/dt = (\mu_0 N^2 A / l) di/dt = L di/dt$

L is called the inductance unit is henry (H). $L = \mu_0 N^2 A / l$

Energy per unit volume inside a solenoid is $B^2 / (2\mu_0)$ (compare to capacitors)
What is the energy inside a solenoid = $(A l) B^2 / (2\mu_0)$

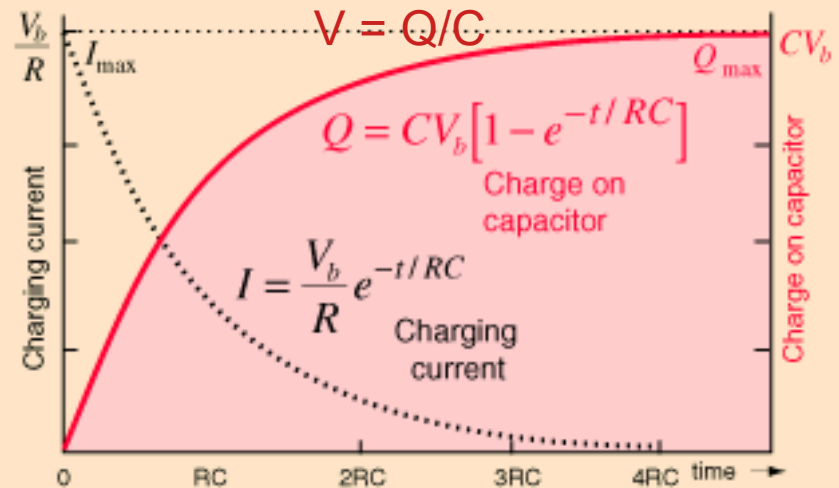


<https://nationalmaglab.org/magnet-academy/watch-play/interactive-tutorials/electromotive-force-in-inductors/>



$t=0$
 $V = \text{emf}, I = 0$

$t = \infty$
 $\text{emf} = 0$
 $I = I_{max} = V/R$



At $t = 0$
 $Q = 0$
 $V_C = 0$
 $I = \frac{V_b}{R}$

As $t \rightarrow \infty$
 $Q \rightarrow CV_b$
 $V_C \rightarrow V_b$
 $I \rightarrow 0$

https://www.electronics-tutorials.ws/rc/rc_1.html

<https://www.electronics-tutorials.ws/inductor/lr-circuits.html>

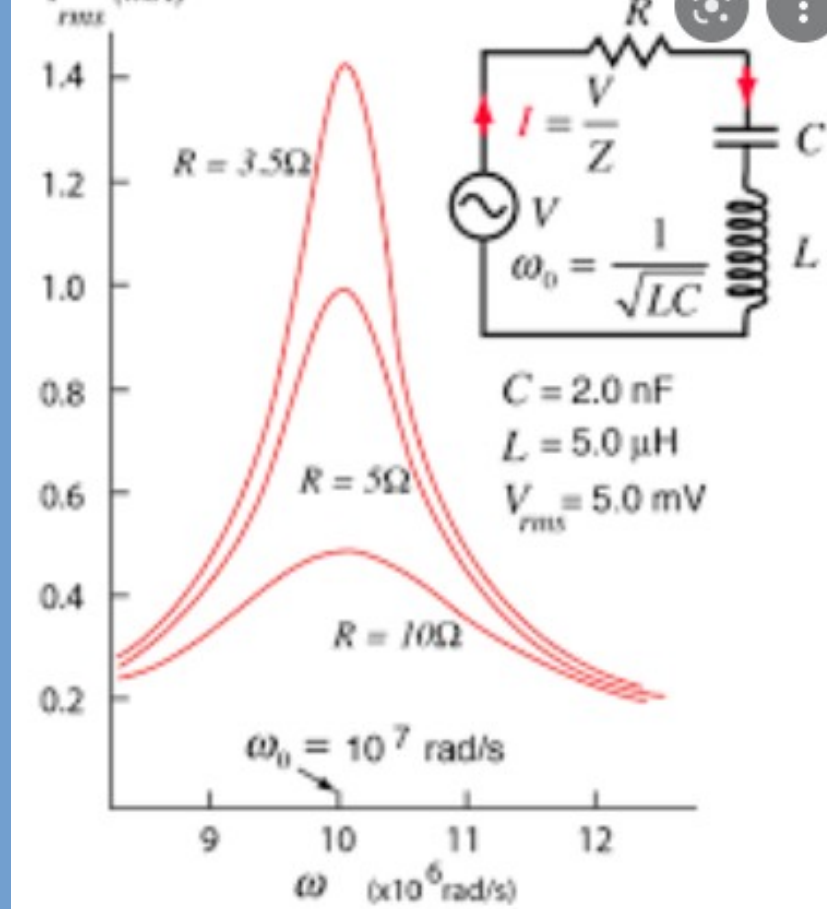
In DC circuits → charge
 Of capacitors/inductors

<https://www.electronics-tutorials.ws/accircuits/ac-inductance.html>

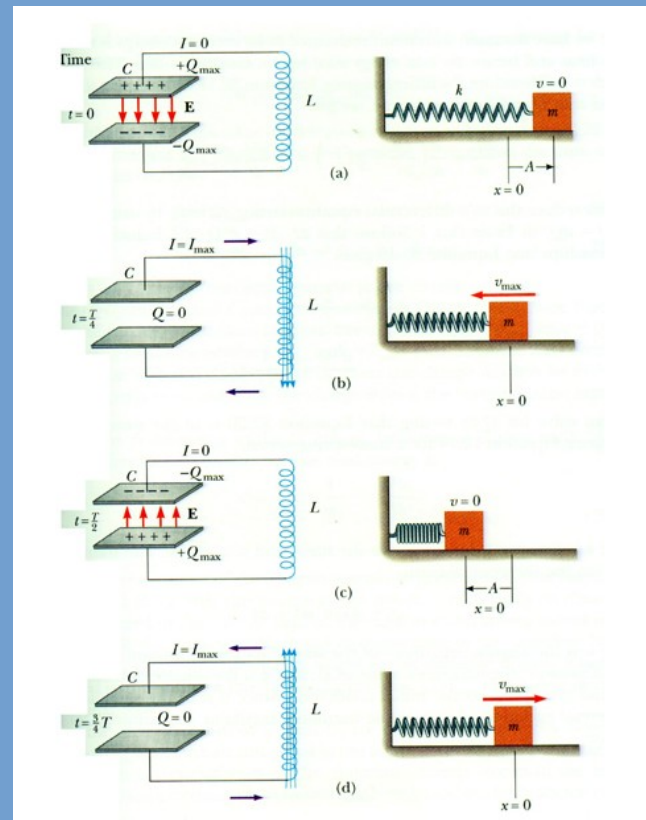
In AC circuits
 Without R

<https://www.electronics-tutorials.ws/accircuits/ac-capacitance.html>

Show the excel file. RL AC circuit



Driven harmonic oscillators



Driven oscillation and resonance. If you push a mass (inertia) on a swing. Inertia resists the change in speed (elephant vs monkey). Gravity is the restoring force. It Brings back the mass. Energy is $PE \rightarrow KE \rightarrow PE..$ In RLC, the V_s drives the oscillation. The inductor resists change in current. Can't change fast. And the capacitor is the restoring force. The charges turn around

<https://www.ux1.eiu.edu/~cfadd/1360/32Ind/LC.html>

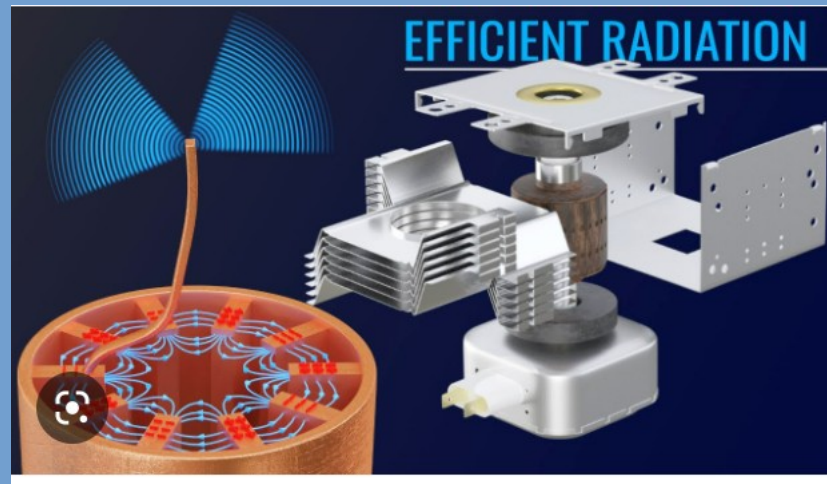
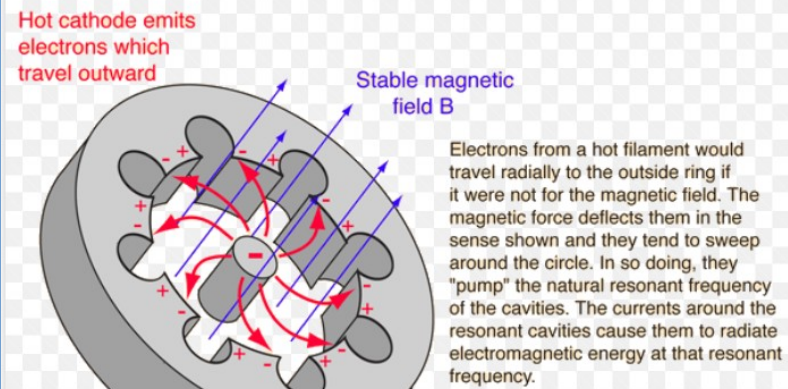
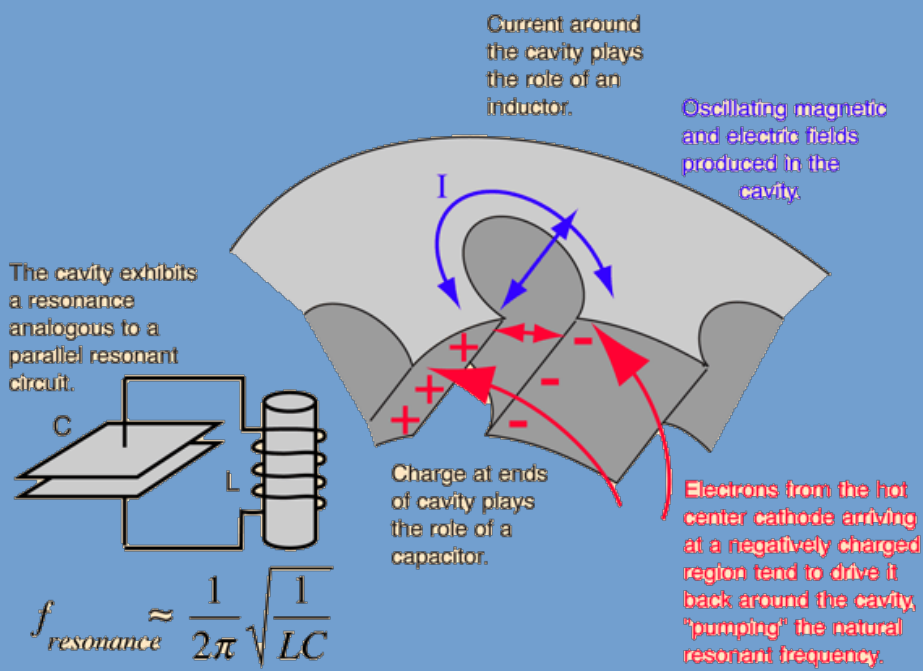
<http://hyperphysics.phy-astr.gsu.edu/hbase/electric/serres.html>

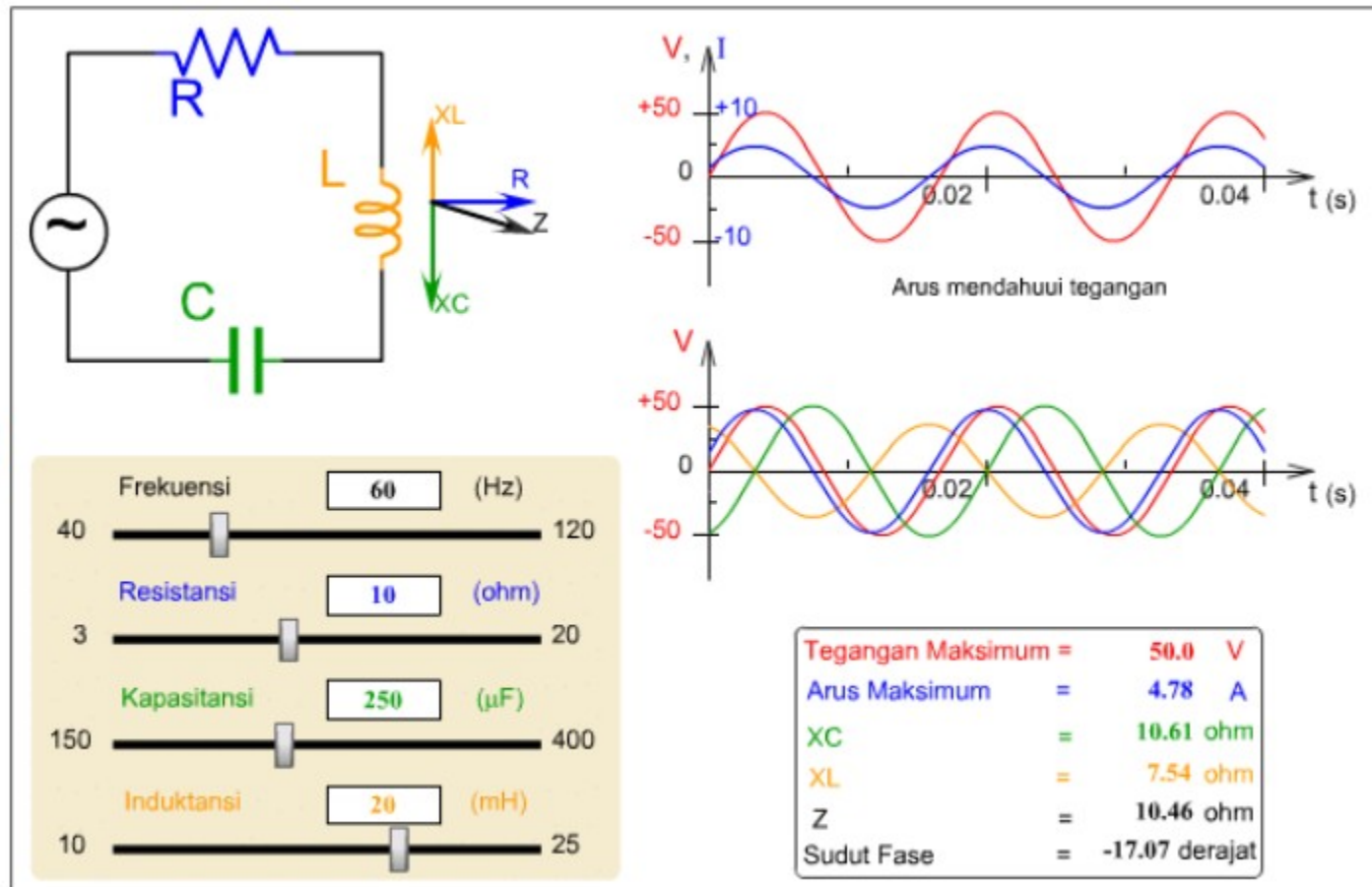


The magnetrons used in microwave ovens tend to have an exit frequency of 2450 MHz and provide an electromagnetic energy of 600 W for an electrical entry power of 1200 W



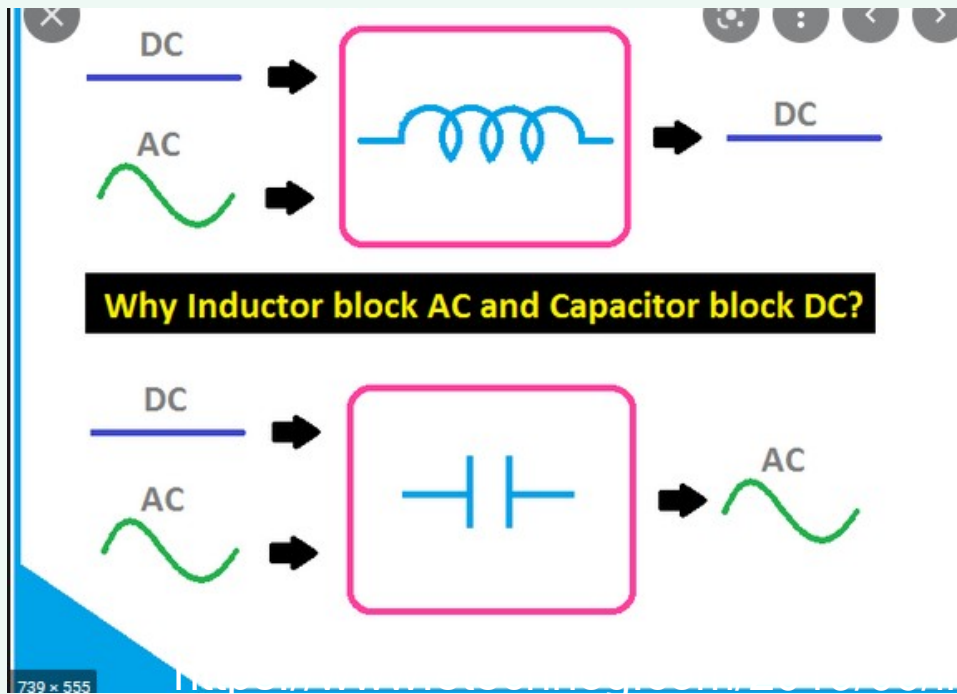
Cavity resonance.





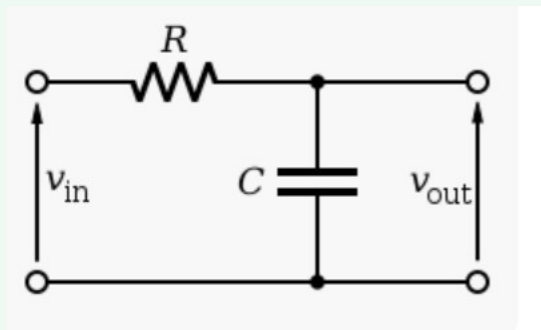
https://javalab.org/en/rlc_serial_circuit_en/

Excellent simulation. Show that I is lagging V_L and I is leading V_C



https://javalab.org/en/rlc_serial_circuit_en/

Show that for high frequency, the circuit is inductive. The capacitor is like a wire.
For low frequency, the circuit is capacitive. The inductor is like a wire.



Low pass filter The high frequencies are dumped through the capacitor.

<https://www.etchnog.com/2019/06/inductor-capacitor-block-ac-dc.html>

22.1 Induced Emf and Induced Current Electromagnetic induction is the phenomenon in which an emf is induced in a piece of wire or a coil of wire with the aid of a magnetic field. The emf is called an induced emf, and any current that results from the emf is called an induced current.

22.2 Motional Emf An emf $\mathcal{E}$ is induced in a conducting rod of length L when the rod moves with a speed v in a magnetic field of magnitude B , according to Equation 22.1, which applies when the velocity of the rod, the length of the rod, and the magnetic field are mutually perpendicular. When the motional emf is used to operate an electrical device, such as a light bulb, the energy delivered to the device originates in the work done to move the rod, and the law of conservation of energy applies.

$$\mathcal{E} = vBL \quad (22.1)$$

22.3 Magnetic Flux The magnetic flux Φ that passes through a surface is given by Equation 22.2, where B is the magnitude of the magnetic field, A is the area of the surface, and ϕ is the angle between the field and the normal to the surface. The magnetic flux is proportional to the number of magnetic field lines that pass through the surface.

$$\Phi = BA \cos \phi \quad (22.2)$$

22.4 Faraday's Law of Electromagnetic Induction Faraday's law of electromagnetic induction states that the average emf $\mathcal{E}$ induced in a coil of N loops is given by Equation 22.3, where $\Delta\Phi$ is the change in magnetic flux through one loop and Δt is the time interval during which the change occurs. Motional emf is a special case of induced emf.

$$\mathcal{E} = -N \left(\frac{\Phi - \Phi_0}{t - t_0} \right) = -N \frac{\Delta\Phi}{\Delta t} \quad (22.3)$$

22.5 Lenz's Law Lenz's law provides a way to determine the polarity of an induced emf. Lenz's law is stated as follows: The induced emf resulting from a changing magnetic flux has a polarity that leads to an induced current whose direction is such that the induced magnetic field opposes the original flux change. This statement is a consequence of the law of conservation of energy.

22.7 The Electric Generator In its simplest form, an electric generator consists of a coil of N loops that rotates in a uniform magnetic field $\vec{B}$. The emf produced by this generator is given by Equation 22.4, where A is the area of the coil, ω is the angular speed (in rad/s) of the coil, and $\mathcal{E}_0 = NAB\omega$ is the peak emf. The angular speed in rad/s is related to the frequency f in cycles/s, or Hz, according to $\omega = 2\pi f$.

$$\mathcal{E} = NAB\omega \sin \omega t = \mathcal{E}_0 \sin \omega t \quad (22.4)$$

When an electric motor is running, it exhibits a generator-like behavior by producing an induced emf, called the back emf. The current I needed to keep the motor running at a constant speed is given by Equation 22.5, where V is the emf applied to the motor by an external source, $\mathcal{E}$ is the back emf, and R is the resistance of the motor coil.

$$I = \frac{V - \mathcal{E}}{R} \quad (22.5)$$

22.8 Mutual Inductance and Self-Inductance Mutual induction is the effect in which a changing current in the primary coil induces an emf in the secondary coil. The average emf $\mathcal{E}_s$ induced in the secondary coil by a change in current ΔI_p in the primary coil is given by Equation 22.7, where Δt is the time interval during which the change occurs. The constant M is the mutual inductance between the two coils and is measured in henries (H).

$$\mathcal{E}_s = -M \frac{\Delta I_p}{\Delta t} \quad (22.7)$$

Self-induction is the effect in which a change in current ΔI in a coil induces an average emf $\mathcal{E}$ in the same coil, according to Equation 22.9. The constant L is the self-inductance, or inductance, of the coil and is measured in henries.

$$\mathcal{E} = -L \frac{\Delta I}{\Delta t} \quad (22.9)$$

To establish a current I in an inductor, work must be done by an external agent. This work is stored as energy in the inductor, the amount of energy being given by Equation 22.10. The energy stored in an inductor can be regarded as being stored in its magnetic field. At any point in air or vacuum or in a nonmagnetic material where a magnetic field $\vec{B}$ exists, the energy density, or the energy stored per unit volume, is given by Equation 22.11.

$$\text{Energy} = \frac{1}{2} LI^2 \quad (22.10)$$

$$\text{Energy density} = \frac{1}{2\mu_0} B^2 \quad (22.11)$$

22.9 Transformers A transformer consists of a primary coil of N_p turns and a secondary coil of N_s turns. If the resistances of the coils are negligible, the voltage V_p across the primary coil and the voltage V_s across the secondary coil are related according to the transformer equation, which is Equation 22.12, where the ratio N_s/N_p is called the turns ratio of the transformer. A transformer functions with ac electricity, not with dc. If the transformer is 100% efficient in transferring power from the primary to the secondary coil, the ratio of the secondary current I_s to the primary current I_p is given by Equation 22.13.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \quad (22.12)$$

$$\frac{I_s}{I_p} = \frac{N_p}{N_s} \quad (22.13)$$